Benchmarking Deep Reinforcement Learning for Navigation in Denied Sensor Environments

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Abstract

Deep Reinforcement learning (DRL) is used to enable autonomous navigation in unknown environments. Most research assume perfect sensor data, but real-world environments may contain natural and artificial sensor noise and denial. Here, we present a benchmark of both well-used and emerging DRL algorithms in a navigation task with configurable sensor denial effects. In particular, we are interested in comparing how different DRL methods (e.g. model-free PPO vs. model-based DreamerV3) are affected by sensor denial. We show that DreamerV3 outperforms other methods in the visual end-to-end navigation task with a dynamic goal - and other methods are not able to learn this. Furthermore, DreamerV3 generally outperforms other methods in sensor-denied environments. In order to improve robustness, we use adversarial training and demonstrate an improved performance in denied environments, although this generally comes with a performance cost on the vanilla environments. We anticipate this benchmark of different DRL methods and the usage of adversarial training to be a starting point for the development of more elaborate navigation strategies that are capable of dealing with uncertain and denied sensor readings.

Keywords: Autonomy; Navigation; Sensor Fusion; Deep Reinforcement Learning; Machine Learning; Complex Environments; Sensor Faults; Sensor Failure; Adversarial Attacks; Adversarial Training; Adversarial Environmental Perturbations

在传感器被拒绝的环境中对深度强化学习进行导航基准测试

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Abstract

深度强化学习（DRL）用于在未知环境中实现自主导航。大多数研究都假设传感器数据是完美的，但现实世界的环境可能包含自然和人为的传感器噪声和否认。在这里，我们提出了导航任务中常用和新兴 DRL 算法的基准，具有可配置的传感器拒绝效果。特别是，我们有兴趣比较不同的 DRL 方法（例如无模型 PPO 与基于模型的 DreamerV3）如何受到传感器拒绝的影响。我们表明，DreamerV3 在具有动态目标的视觉端到端导航任务中优于其他方法 - 而其他方法无法学习这一点。此外，在没有传感器的环境中，DreamerV3 通常优于其他方法。为了提高鲁棒性，我们使用对抗性训练并在拒绝环境中展示了改进的性能，尽管这通常会在普通环境中带来性能成本。我们预计不同 DRL 方法的基准和对抗性训练的使用将成为开发更精细的导航策略的起点，这些策略能够处理不确定和否认的传感器读数。

Keywords: 自治;导航;传感器融合；深度强化学习；机器学习；复杂的环境；传感器故障；传感器故障；对抗性攻击；对抗训练；对抗性环境扰动

1 Introduction

Autonomous navigation is a fundamental challenge in unmanned systems. Conventional autonomous systems typically use Simultaneous Localisation and Mapping (SLAM) alongside Kalman filters that fuse data from multiple sources to map the environment, localize the agent, and generate trajectories to the goal. In recent years, deep reinforcement learning (DRL) methods have been used to train end-to-end policies that can navigate using a camera. These improvements have been made aided by advances in digital image processing largely due to convolutional neural networks trained using backpropagation [1–3], and parametrization of RL using deep neural networks [4].

Whilst autonomous visual navigation is achievable in simulated environments [5], most of the research assumes perfect sensor readings. But, in real-world scenarios, information about the environment may be uncertain, incomplete, or even intentionally denied. This can limit the ability of the navigation system to adapt to changes in the environment and make optimal decisions. These failure modes and how they affect the systems need to be understood for safety-critical operations, such as unmanned aerial vehicles or autonomous road vehicles. In the case of DRL agents, which are notoriously hard to explain, it is important to study how the sensor artefacts or failures affect the training and evaluation.

Further, real systems typically use a suite of sensors. Whilst a lot of the cutting-edge work in DRL focuses on end-to-end visual navigation, in practice it is important to understand how the models learn across different modalities, by taking in data from multiple sensors. In this work, we study how sensor failure affects the training and evaluation of policies that use a camera or Lidar as the observation space.

To understand the effects of sensor noise and failures, we compare several DRL architectures: TD3, PPO, PPO-LSTM, and DreamerV3. TD3 and PPO are a commonly used for evaluating environments with continuous action spaces, PPO-LSTM is the recurrent version of PPO (recurrent models were shown to outperform non-recurrent counterparts by Mirowski et al. [6]), and DreamerV3 which uses autoencoders to create an internal world model of the environment, and was shown to outperform PPO on certain environments.

We generate a baseline of their performance in a 'vanilla' environment, with no perturbations - noise or failures - to the sensors. We then progressively add random areas to the environment in which the sensors are perturbed. The policies are then evaluated to understand how successful they are in navigation under varying degrees of noise. To train and evaluate the methods we use a modified version of the ROS-based gymnasium environment for training navigation policies, *DRL-Robot-Navigation* [7]. It contains a 3D maze in which a robot is tasked with navigating to a goal. This publication is an extension of the conference paper on autonomous navigation presented at ICUAS 2024 [8].

1 Introduction

自主导航是无人系统的基本挑战。传统的自主系统通常使用同步定位和建图（SLAM）以及卡尔曼滤波器，融合来自多个源的数据来绘制环境图、定位代理并生成到达目标的轨迹。近年来，深度强化学习（DRL）方法已被用来训练可以使用摄像头进行导航的端到端策略。这些改进得益于数字图像处理的进步，这主要归功于使用反向传播 [1-3] 训练的卷积神经网络，以及使用深度神经网络 [4] 的强化学习参数化。

虽然自主视觉导航可以在模拟环境中实现 [5]，但大多数研究都假设传感器读数完美。但是，在现实场景中，有关环境的信息可能是不确定的、不完整的，甚至是故意否认的。这会限制导航系统适应环境变化并做出最佳决策的能力。对于安全关键型操作（例如无人机或自动道路车辆），需要了解这些故障模式及其对系统的影响。对于众所周知难以解释的 DRL 代理来说，研究传感器伪影或故障如何影响训练和评估非常重要。

此外，实际系统通常使用一套传感器。虽然 DRL 的许多前沿工作都集中在端到端视觉导航上，但在实践中，了解模型如何通过从多个传感器获取数据来跨不同模式进行学习非常重要。在这项工作中，我们研究传感器故障如何影响使用相机或激光雷达作为观察空间的策略的训练和评估。

为了了解传感器噪声和故障的影响，我们比较了几种 DRL 架构：TD3、PPO、PPO-LSTM 和 DreamerV3。 TD3 和 PPO 常用于评估具有连续动作空间的环境，PPO-LSTM 是 PPO 的循环版本（Mirowski 等人 [6] 证明循环模型优于非循环模型），而 DreamerV3 使用自动编码器创建环境的内部世界模型，并在某些环境中表现优于 PPO。

我们在“普通”环境中生成了它们的性能基线，传感器没有受到干扰（噪音或故障）。然后，我们逐步将随机区域添加到传感器受到干扰的环境中。然后对这些策略进行评估，以了解它们在不同程度的噪音下导航的成功程度。为了训练和评估这些方法，我们使用基于 ROS 的体育馆环境的修改版本来训练导航策略，*DRL-Robot-Navigation* [7]。它包含一个 3D 迷宫，其中机器人的任务是导航到目标。本出版物是 ICUAS 2024 上发表的自主导航会议论文的延伸[8]。

1.1 Review of DRL Applications to Navigation

Several studies have achieved end-to-end navigation using DRL with the camera as the sole sensor in a simulation environment [9]. Popular environments include the DeepMind Lab [10], ViZDoom [11], and DRL-Robot-Navigation [7].

Mirowski et al. [6] present an application of the A3C [12] algorithm to the Deepmind Lab environment. They use the camera as the sole sensor and their best-performing architecture consists of the image encoder, 2 LSTM layers, depth and loop predictions. The agent is rewarded for reaching the goal, or subgoals in the form of a fruit. The agent is not penalized for wall collisions, allowing strategies such as 'wall-following' that do not require memory. DRL can also be used to race FPV unmanned aerial vehicles (UAVs) against expert human pilots [13] by using the image sensor combined with the IMU (which are combined to detect gates and generate VIO states input to the DRL algorithm) in static environments. Polvara et al. show how a hierarchy of DQNs can be used for the autonomous landing of UAVs [14]. Their method handles a large variety of simulated environments and outperforms human pilots in some conditions. Interpretable deep reinforcement learning methods for end-to-end autonomous driving can handle complex urban scenarios [15]. Their method outperforms many baselines including Deep Q-Networks (DQNs), deep deterministic policy gradient (DDPG), Twin Delayed DDPG (TD3), and Soft Actor Critic (SAC) in urban scenarios with surrounding vehicles. The memory of FPV agents in 3D mazes is evaluated by Pasukonis et al. [16]. They find that models such as Dreamer and IMPALA can match humans on smaller mazes (9x9), but their performance significantly drops off on larger mazes (15x15). Lample and Chaplot [17] show that DRL can be used to train agents to play the first-person-shooter game DOOM, in which the agent is required to navigate a 3D maze, collect items, and shoot at enemies. They found that by using DQN and DQRN with an LSTM layer, the agent can converge, but required an addition of predicting game features (using a separate loss function) to be able to reliably shoot at enemies. Liquid neural networks [18] have been proposed in recent years to achieve robust navigation for changing environments [19]. In most of these cases, whilst the physical environment is unknown or stochastic, the papers did not consider compromised sensor readings (bar liquid neural networks that consider noise and other perturbations).

1.2 Review of Sensor Fusion under Attacks

1.2.1 Trajectory Estimation Approaches

The first approach deals with intermittent attacks or denial, for example when GNSS coverage is lost and a Kalman filter (or more advanced recurrent neural network) is used to hold over the trajectory estimation until coverage recovers [20, 21]. This is useful for brief outages. The same idea can preemptively discover other V2V networks to improve localisation [22]. However, if the environment requires constant sharp manoeuvring to avoid dynamic objects, this scenario rules out trajectory estimation approaches. Also, if the attacks are continuous and erode the holdover accuracy, this would be unsuitable.

1.1 Review of DRL Applications to Navigation

一些研究已经在模拟环境中使用 DRL 并以摄像头作为唯一传感器实现了端到端导航 [9]。流行的环境包括 DeepMind Lab [10]、ViZDoom [11] 和 DRL-Robot-Navigation [7]。

米罗斯基等人。 [6] 展示了 A3C [12] 算法在 Deepmind Lab 环境中的应用。他们使用相机作为唯一的传感器，其性能最佳的架构由图像编码器、2 个 LSTM 层、深度和循环预测组成。智能体因达到目标或子目标而获得水果形式的奖励。智能体不会因墙壁碰撞而受到惩罚，从而允许采取不需要记忆的“墙壁跟随”等策略。 DRL 还可用于在静态环境中将图像传感器与 IMU（组合起来检测门并生成输入到 DRL 算法的 VIO 状态输入）相结合，让 FPV 无人机 (UAV) 与专业人类飞行员进行竞赛 [13]。波尔瓦拉等人。展示了如何使用 DQN 层次结构来实现无人机的自主着陆 [14]。他们的方法可以处理各种各样的模拟环境，并且在某些条件下优于人类飞行员。用于端到端自动驾驶的可解释的深度强化学习方法可以处理复杂的城市场景[15]。他们的方法优于许多基线，包括在有周围车辆的城市场景中的深度 Q 网络 (DQN)、深度确定性策略梯度 (DDPG)、双延迟 DDPG (TD3) 和 Soft Actor Critic (SAC)。 Pasukonis 等人评估了 3D 迷宫中 FPV 智能体的记忆。 [16]。他们发现，Dreamer 和 IMPALA 等模型可以在较小的迷宫 (9x9) 上与人类相匹配，但它们在较大的迷宫 (15x15) 上的表现会显着下降。 Lample 和 Chaplot [17] 表明，DRL 可用于训练智能体玩第一人称射击游戏《DOOM》，其中智能体需要导航 3D 迷宫、收集物品并向敌人射击。他们发现，通过将 DQN 和 DQRN 与 LSTM 层结合使用，代理可以收敛，但需要添加预测游戏功能（使用单独的损失函数）才能可靠地射击敌人。近年来，人们提出了液体神经网络[18]来实现针对不断变化的环境的鲁棒导航[19]。在大多数情况下，虽然物理环境未知或随机，但论文并未考虑受损的传感器读数（考虑噪声和其他扰动的液体神经网络除外）。

1.2 Review of Sensor Fusion under Attacks

1.2.1 Trajectory Estimation Approaches

第一种方法处理间歇性攻击或拒绝，例如当 GNSS 覆盖范围丢失时，使用卡尔曼滤波器（或更先进的循环神经网络）来保持轨迹估计，直到覆盖范围恢复 [20, 21]。这对于短暂的中断很有用。同样的想法可以先发制人地发现其他V2V网络以改善本地化[22]。然而，如果环境需要持续的急剧机动以避免动态物体，则这种情况排除了轨迹估计方法。此外，如果攻击是连续的并削弱了保持精度，则这将是不合适的。

1.2.2 Sensor Fusion and Generative AI Approaches

The alternative is suitable for high-end platforms that can afford a diverse sensor suite, performing sensor fusion to alleviate the impact of one sensor being compromised. This is usually unsuitable for low-end drones and does not guarantee success [23]. A cheaper alternative is to use generative AI to generate alternative sensor representations using a single cheap sensor. For example, infrared images can be generated using a camera sensor and then fused together to achieve more robust recognition of the environment [24]. However, these approaches require an onboard diverse sensor suite or a powerful GAN operating onboard, which does not guarantee success.

1.2.3 Adversarial Environmental Perturbations in DRL

First-person-view (FPV) navigation problems are typically considered partially observable [25] because only a portion of the entire state is visible to the agent at any given time. But, sensor faults also have to be considered. Robust partially observable MDP [26] considers the POMDP to have uncertain parameters. Other approaches [27] consider the interaction between the adversary and the agent to be a zero-sum minmax game and define it as a Probabilistic Action Robust MDP. Zhang et al. [28] consider the addition of adversarial perturbations as a state-adversarial MDP (SA-MDP). In this framing, the state returned by the environment may be perturbed by an adversary (an adversary may be another agent or an environmental process that perturbs the state observations), making the observation uncertain and resulting in the action returned by the agent being potentially sub-optimal. Because we consider that the adversary can be static (e.g. it is part of the environment), we prefer the SA-MDP framing for our problem.

When building robust RL models, some of these works focus on physical perturbations [29] that may affect the physics of the agent (e.g. by introducing uncertain forces), whilst others consider perturbations to the sensor state [30]. Korkmaz [31, 32] argues that vanilla training results in more robust policies compared with state-of-the-art adversarial and robust training techniques. Havens et al. [33] propose a hierarchical RL method of switching sub-policies in the presence of adversaries. Applications of adversarial training [34] show how a zero-sum action-robust RL method can be used to reduce conflicts in air mobility traffic management scenarios.

1.3 Gap and Novelty

Although research on DRL and sensor attacks exists, we have not found elaborate studies combining the effect of sensor failure on DRL performance during learning navigation policies. We aim to understand the effects of sensor failures - a potentially disastrous failure mode in autonomous systems - on the ability of DRL agents to learn navigation policies. To do this we:

- Benchmark different reinforcement learning algorithms on the *DRL-Robot-Navigation* environment.
- Study the effect of observation spaces - Lidar and camera - on the navigation capabilities.

1.2.2 Sensor Fusion and Generative AI Approaches

另一种选择适用于能够提供多种传感器套件的高端平台，执行传感器融合以减轻一个传感器受到损害的影响。这通常不适合低端无人机，并且不能保证成功[23]。一种更便宜的替代方案是使用生成式人工智能，使用单个廉价传感器生成替代传感器表示。例如，可以使用相机传感器生成红外图像，然后将其融合在一起以实现更鲁棒的环境识别[24]。然而，这些方法需要机载多样化传感器套件或机载运行的强大 GAN，这并不能保证成功。

1.2.3 Adversarial Environmental Perturbations in DRL

第一人称视角 (FPV) 导航问题通常被认为是部分可观察的 [25]，因为在任何给定时间，代理只能看到整个状态的一部分。但是，还必须考虑传感器故障。鲁棒部分可观测 MDP [26] 认为 POMDP 具有不确定的参数。其他方法 [27] 将对手和代理之间的交互视为零和最小最大博弈，并将其定义为概率动作鲁棒 MDP。张等人。 [28]考虑添加对抗性扰动作为状态对抗性 MDP (SA-MDP)。在这个框架中，环境返回的状态可能会受到对手的干扰（对手可能是另一个代理或扰乱状态观察的环境过程），使得观察不确定并导致代理返回的动作可能不是最优的。因为我们认为对手可以是静态的（例如，它是环境的一部分），所以我们更喜欢使用 SA-MDP 框架来解决我们的问题。

在构建鲁棒的强化学习模型时，其中一些工作关注可能影响代理物理特性的物理扰动[29]（例如通过引入不确定的力），而其他工作则考虑对传感器状态的扰动[30]。 Korkmaz [31, 32] 认为，与最先进的对抗性和稳健的训练技术相比，普通训练会产生更稳健的策略。哈文斯等人。 [33]提出了一种在对手存在的情况下切换子策略的分层强化学习方法。对抗性训练的应用[34]展示了如何使用零和动作鲁棒强化学习方法来减少空中交通管理场景中的冲突。

1.3 Gap and Novelty

尽管存在关于 DRL 和传感器攻击的研究，但我们还没有发现在学习导航策略期间结合传感器故障对 DRL 性能影响的详细研究。我们的目标是了解传感器故障（自治系统中一种潜在的灾难性故障模式）对 DRL 代理学习导航策略的能力的影响。为此，我们:

- 在 *DRL-Robot- Navigation* 环境上对不同的强化学习算法进行基准测试。
- 研究观测空间（激光雷达和摄像头）对导航能力的影响。

- Perform quantitative and qualitative studies showing the effects of adversarial perturbations - noise and sensor failure - on the performance of DRL algorithms in the navigation task.

Further, to enable others to replicate our work, we open source the modifications to the *DRL-Robot-Navigation* environment, and update it to fit the gymnasium [35] interface.[1]

In this paper, we explain the methodology, i.e. the environment, changes to the environment, and the choice of reinforcement learning models, in section 2. We present the results of the benchmark of different algorithms on the environment, the Lidar and camera noise studies, and discussion with respect to literature in section 3. Finally, we conclude and make suggestions for further work in section 4.

2 Methodology

We build on top of the *DRL-Robot-Navigation* environment [7] which simulates a robot navigating around a 10 m x 10 m maze, populated with static obstacles. The robot has a suite of sensors - Lidar, camera, and odometry - but only a subset of these was used as the state observation to train the reinforcement learning model in the original paper. The original paper uses Lidar, distance to goal, and angle to goal, concatenated together as the observation space, and the camera is not used. An illustration of the environment, which shows the *DRL-Robot-Navigation* environment along with sensor readings, is presented in Fig. 1.

As the focus of this study is on studying adversarial sensor perturbations and testing different sensor suites, the changes to the environment are described in section 2.1. The benchmarking procedure, of training the different RL algorithms - TD3, PPO, PPO LSTM, and DreamerV3, using different modalities and with differing sensor noise, is illustrated in figure 2. The reinforcement learning algorithms used in the benchmark are described in section 2.2. Training experimental settings, including hyperparameters, are shown in section 2.3. The evaluation process is described in section 2.4.

2.1 Changes to the DRL-Robot-Navigation Environment

To enable our experiments, the *DRL-Robot-Navigation* environment required several changes. A primary change to the environment is the addition of a vision-based observation space. To do this, the image from the robot is outputted from the environment, and a visual marker is added to mark the position of the goal. A pink sphere of diameter 0.5 m is added where the goal resides and is shown in Fig. 1 (top left image).

The collision system is updated, as in the original implementation, it relied on the reading from the Lidar sensor. This is not a reliable way of detecting collisions, and adding noise to the Lidar sensor could trigger collisions. Instead, we generate a flag and detect any physical collisions between objects by triggering ROS topic that can be subscribed.

[1] https://github.com/mazqtpopx/cranfield-navigation-gym

- 进行定量和定性研究，显示对抗性扰动（噪声和传感器故障）对导航任务中 DRL 算法性能的影响。

此外，为了让其他人能够复制我们的工作，我们开源了对 *DRL-Robot-Navigation* 环境的修改，并更新它以适应体育馆 [35] 界面。[1]

在本文中，我们在第 2 节中解释了方法，即环境、环境的变化和强化学习模型的选择。我们在第 3 节中介绍了不同算法对环境的基准测试结果、激光雷达和相机噪声研究，以及关于文献的讨论。最后，我们在第 4 节中总结并提出了进一步工作的建议。

2 Methodology

我们构建在 *DRL-Robot-Navigation* 环境 [7] 之上，该环境模拟机器人在 10 m x 10 m 的迷宫中导航，迷宫中充满了静态障碍物。该机器人拥有一套传感器——激光雷达、摄像头和里程计——但在原始论文中，只有其中的一个子集被用作状态观察来训练强化学习模型。原论文使用激光雷达、到目标的距离和到目标的角度，串联在一起作为观察空间，并且没有使用相机。图 1 展示了环境图示，其中显示了 *DRL-Robot-Navigation* 环境以及传感器读数。

由于本研究的重点是研究对抗性传感器扰动并测试不同的传感器套件，因此环境的变化在第 2.1 节中进行了描述。图 2 说明了使用不同模态和不同传感器噪声训练不同 RL 算法（TD3、PPO、PPO LSTM 和 DreamerV3）的基准测试程序。基准测试中使用的强化学习算法在第 2.2 节中描述。训练实验设置，包括超参数，如第 2.3 节所示。评估过程如第 2.4 节所述。

2.1 Changes to the DRL-Robot-Navigation Environment

为了进行我们的实验，*DRL-Robot-Navigation* 环境需要进行一些更改。环境的主要变化是增加了基于视觉的观察空间。为此，从环境中输出机器人的图像，并添加视觉标记来标记目标的位置。在球门所在位置添加一个直径为 0.5 m 的粉色球体，如图 1（左上图）所示。

碰撞系统已更新，与最初的实现一样，它依赖于激光雷达传感器的读数。这不是检测碰撞的可靠方法，并且向激光雷达传感器添加噪声可能会触发碰撞。相反，我们生成一个标志，并通过触发可以订阅的 ROS 主题来检测对象之间的任何物理碰撞。

[1] https://github.com/mazqtpopx/cranfield-navigation-gym

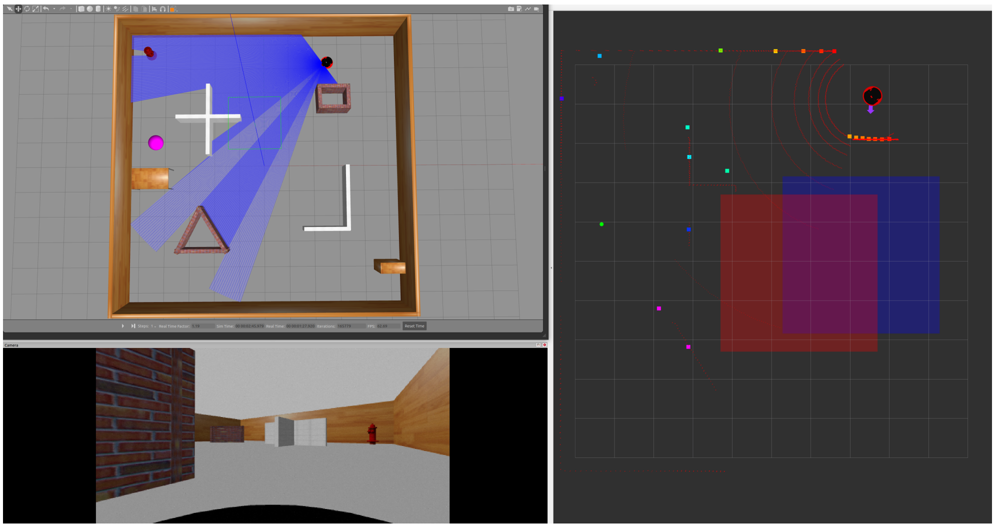


Fig. 1 Example of the *DRL-Robot-Nav Maze* Environment [7]. Top-left: top-down view of the environment rendered in Gazebo. Bottom-left: the view from the RGB camera. Right: RViz visualization of the discretized Lidar sensor readings, the robot, and the noisy sensor areas (blue: camera noise area, red: Lidar noise area). The pink sphere (top left) is added as a visual cue to mark the goal for vision-based policies.

We added areas that contain sensor noise: separately for Lidar and camera. The areas are fixed-size rectangles that spawn at a random point of the map at the start of each episode. The way the sensor faults are modelled is described further in section 2.1.1.

We updated the reward system. Originally, the agent received 100 reward for reaching the goal and -100 for a collision with a wall. Otherwise, a function of the actions (penalizing the robot for inaction) and the minimum distance from a wall (based on the Lidar observation, if the robot is closer than 1 m to a wall) was returned. First, we normalized the reward between +1 and -1. Next, we added a penalty for taking a step: of size -1 divided by the maximum number of steps. This was done to prevent a local minima where the robot doesn't move and is in line with other gymnasium environments [35]. We also considered the function of penalizing inaction and Lidar distance to the wall to be unnecessary. Instead, we consider that these items - inaction and obstacle avoidance - should be learned on the policy side from observation without reward shaping.

2.1.1 Modelling Adversarial Sensor Perturbations

Autonomous vehicles often use a camera as a primary sensor for navigation, but physical sensor attacks are rarely considered. Kim et al. [36] perform a review of drone attacks, including laser attacks on the camera sensor. They find that when applied, the camera pixels are distorted. This shows that camera sensors are susceptible to

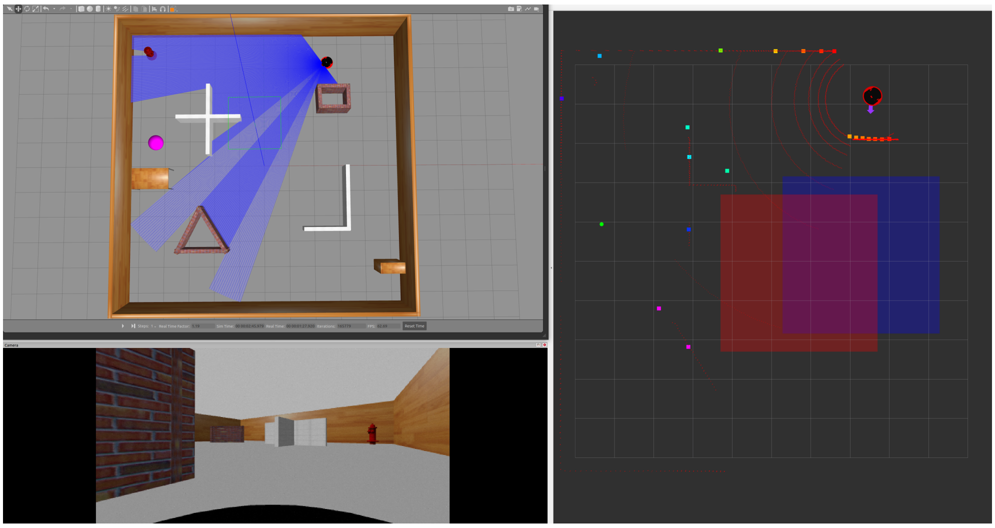


Fig. 1 *DRL-Robot-Nav Maze* 环境示例 [7]。左上：Gazebo 中呈现的环境的自上而下视图。左下：RGB 相机的视图。右：RViz 离散激光雷达传感器读数、机器人和噪声传感器区域的可视化（蓝色：相机噪声区域，红色：激光雷达噪声区域）。添加粉红色球体（左上角）作为视觉提示，以标记基于愿景的政策的目标。

我们添加了包含传感器噪声的区域：分别针对激光雷达和相机。这些区域是固定大小的矩形，在每集开始时在地图的随机点生成。传感器故障的建模方式将在第 2.1.1 节中进一步描述。

我们更新了奖励系统。最初，智能体因达到目标而获得 100 奖励，因与墙壁碰撞而获得 -100 奖励。否则，返回动作（惩罚不动作的机器人）和距墙壁的最小距离（基于激光雷达观测，如果机器人距墙壁小于 1 m）的函数。首先，我们将奖励标准化在 +1 和 -1 之间。接下来，我们添加了一步的惩罚：大小为 -1 除以最大步数。这样做是为了防止机器人不移动并与其他体育馆环境保持一致的局部最小值[35]。我们还认为惩罚不作为和激光雷达距墙距离的功能是不必要的。相反，我们认为这些项目——不作为和避免障碍——应该在政策方面通过观察而不是奖励塑造来学习。

2.1.1 Modelling Adversarial Sensor Perturbations

自动驾驶汽车通常使用摄像头作为导航的主要传感器，但很少考虑物理传感器攻击。金等人。 [36] 对无人机攻击进行审查，包括对相机传感器的激光攻击。他们发现应用后，相机像素会扭曲。这表明相机传感器容易受到

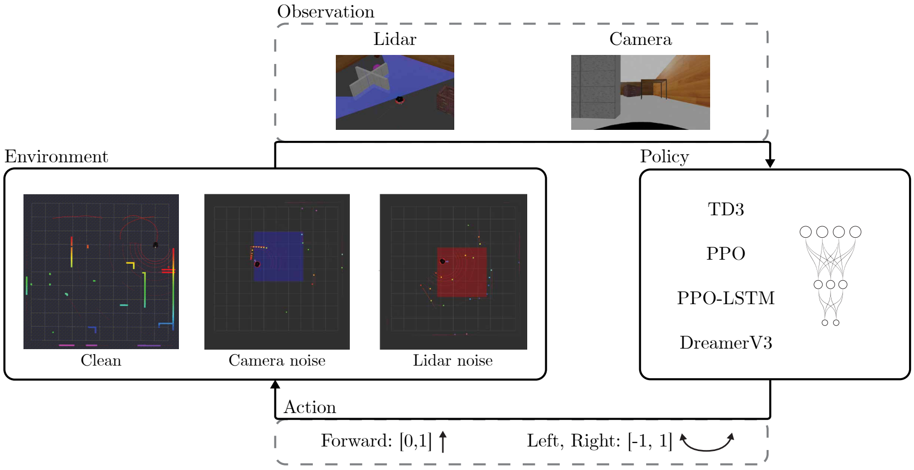
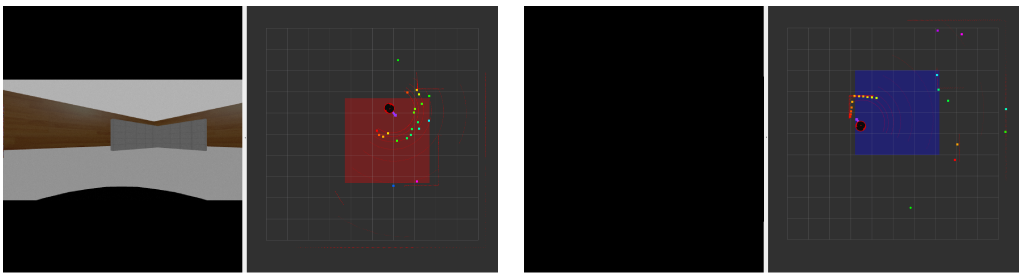


Fig. 2 Experimental diagram showing the process of the presented experiments. The environment consists of clean, camera noise, and Lidar noise variants. The environment outputs the observation in form of two modalities, Lidar and camera. Multiple algorithms are then trained to navigate to the goal whilst avoiding the obstacles.

physical attacks. Camera attacks are modelled as a complete failure, i.e. all the pixels are turned black (0) whilst inside of the camera noise rectangle. Whilst this is not explicitly modelled after a physical attack, it is a possible temporary failure mode for a camera and is shown in Fig. 3(b). Lidar attacks are modelled by adding a strong Gaussian noise, ranging between 0-5 meters to each of the Lidar points. Sensor noise is applied evenly throughout the noisy area. A Lidar sensor perturbation is shown in Fig. 3(a)



(a) (b)

Fig. 3 Example of a robot in the Lidar noise area, visualized by RViz. Once inside the red Lidar noise area, the discretized Lidar points have Gaussian noise added to them and no longer provide accurate representations of the environment. Example of a robot in the camera noise area, visualized by RViz. Once inside the blue camera noise area, the image feed (shown on the left) has all pixels turned to black - this simulates a camera failure.

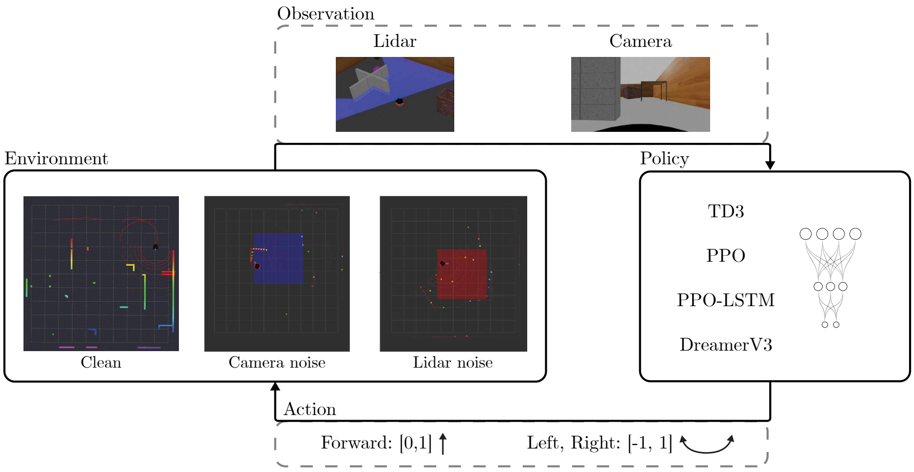
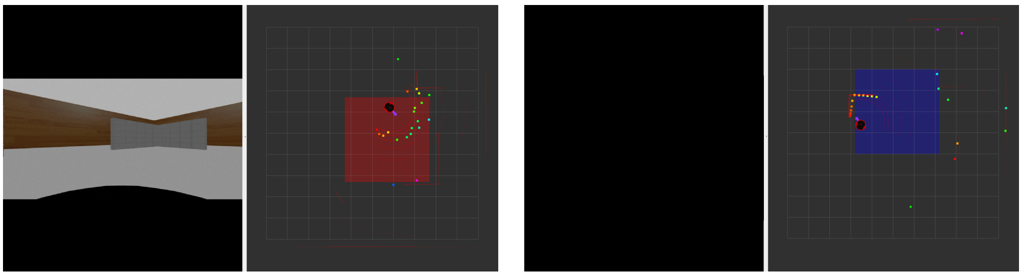


Fig. 2 实验图显示了所提出的实验的过程。环境由干净、相机噪声和激光雷达噪声变体组成。环境以两种方式输出观察结果：激光雷达和摄像头。然后训练多种算法来导航到目标，同时避开障碍物。

物理攻击。相机攻击被建模为完全失败，即相机噪声矩形内部的所有像素都变黑（0）。虽然这没有在物理攻击后明确建模，但它是相机可能的临时故障模式，如图 3(b) 所示。激光雷达攻击通过添加强高斯噪声来建模，范围在每个激光雷达点 0-5 米之间。传感器噪声均匀地施加到整个噪声区域。激光雷达传感器扰动如图 3(a) 所示



(a) (b)

Fig. 3 激光雷达噪声区域中的机器人示例，由 RViz 可视化。一旦进入红色激光雷达噪声区域，离散激光雷达点就会添加高斯噪声，并且不再提供环境的准确表示。相机噪声区域中的机器人示例，由 RViz 可视化。一旦进入蓝色相机噪声区域，图像输入（如左侧所示）的所有像素都变成黑色 - 这模拟了相机故障。

2.2 Reinforcement Learning Benchmark

A benchmark of the following RL methods is performed on the *DRL-Robot-Navigation* environment:

- **Twin Delayed DDPG (TD3)** [37] is an off-policy, model-free algorithm designed to work on continuous action spaces. It employs two Q-value approximators (critics) and uses the smaller of the two Q-values to update the policy, which helps to mitigate the problem of overestimation bias. TD3 also introduces a delay in updating the policy network to reduce the variance of policy updates and adds noise to the target action to encourage exploration. The hyperparameters used to train TD3 are shown in table 1.
- **Proximal Policy Optimization (PPO)** [38] is an on-policy, model-free algorithm designed to work on continuous spaces. **Recurrent PPO**, a modification of PPO with the introduction of a long short-term memory (LSTM) layer [39], is also used. The hyperparameters used to train PPO and Recurrent PPO are shown in table 2. The recurrent architecture is similar to the PPO, but contains an LSTM layer before the Actor-Critic Multilayer Perceptron (but after the convolutional layers for the camera observation).
- **DreamerV3** [40] is an off-policy, model-based algorithm that learns a world model from the experience by incorporating an autoencoder that encodes the input into a discrete representation. It uses recurrence and predicts the dynamics, reward, and the continuity of the episode. The hyperparameters are fixed, hence tuning is not required. It has shown to outperform PPO on several environments, including navigation on DeepMind Lab [10]. Based on this, it is expected that DreamerV3 should outperform the other algorithms on our navigation problems. The default model is used.

Stable-Baselines3 [41] implementations of TD3, PPO, and recurrent PPO were used for the experiments. The general setup for TD3 and PPO is shown in Fig. 4, showing the networks for both camera and Lidar observations. For the camera observation, the image is initially passed through a convolutional network to reduce the dimensionality to 256 features. Those features are then passed into the actor-critic MLP. In the case of the recurrent PPO, the features are first passed through an LSTM layer, before going to the actor-critic. For the Lidar, because it only contains 24 data points, the features are passed straight to the actor-critic. We perform:

- A benchmark of the RL methods.
- A learning-rate discovery to find the optimal LR for each algorithm as it is the most sensitive hyperparameter (apart from Dreamerv3 as it does not require hyperparameter tuning). This is shown in the appendix.
- A study into the performance of the agents across different observation spaces in sensor-perturbed areas.

2.2 Reinforcement Learning Benchmark

以下 RL 方法的基准测试是在 *DRL-Robot-Navigation* 环境上执行的：

- **Twin Delayed DDPG (TD3)** [37]是一种离策略、无模型的算法，设计用于连续动作空间。它采用两个 Q 值逼近器（批评者）并使用两个 Q 值中较小的一个来更新策略，这有助于减轻高估偏差的问题。 TD3 还引入了更新策略网络的延迟，以减少策略更新的方差，并向目标操作添加噪声以鼓励探索。用于训练TD3的超参数如表1所示。

- **Proximal Policy Optimization (PPO)** [38]是一种在策略、无模型算法，设计用于连续空间。还使用了 **Recurrent PPO**，它是 PPO 的修改，引入了长短期记忆（LSTM）层 [39]。用于训练 PPO 和循环 PPO 的超参数如表 2 所示。循环架构与 PPO 类似，但在 Actor-Critic 多层感知器之前包含一个 LSTM 层（但在用于摄像机观察的卷积层之后）。

- **DreamerV3** [40]是一种基于模型的离策略算法，通过结合将输入编码为离散表示的自动编码器，从经验中学习世界模型。它使用循环并预测情节的动态、奖励和连续性。超参数是固定的，因此不需要调整。它已在多种环境中表现优于 PPO，包括 DeepMind Lab 上的导航 [10]。基于此，预计 DreamerV3 在我们的导航问题上应该优于其他算法。使用默认模型。

TD3、PPO 和循环 PPO 的稳定基线 3 [41] 实现用于实验。 TD3 和 PPO 的一般设置如图 4 所示，显示了相机和激光雷达观测的网络。对于相机观察，图像最初通过卷积网络以将维度减少到 256 个特征。然后这些特征会被传递到 actor-critic MLP 中。在循环 PPO 的情况下，特征首先通过 LSTM 层，然后再传递给 actor-critic。对于激光雷达，因为它只包含 24 个数据点，所以特征会直接传递给 actor-critic。我们执行：

- 强化学习方法的基准。
- 学习率发现，为每个算法找到最佳 LR，因为它是最敏感的超参数（Dreamerv3 除外，因为它不需要超参数调整）。这如附录所示。
- 研究传感器扰动区域中不同观察空间中代理的性能。

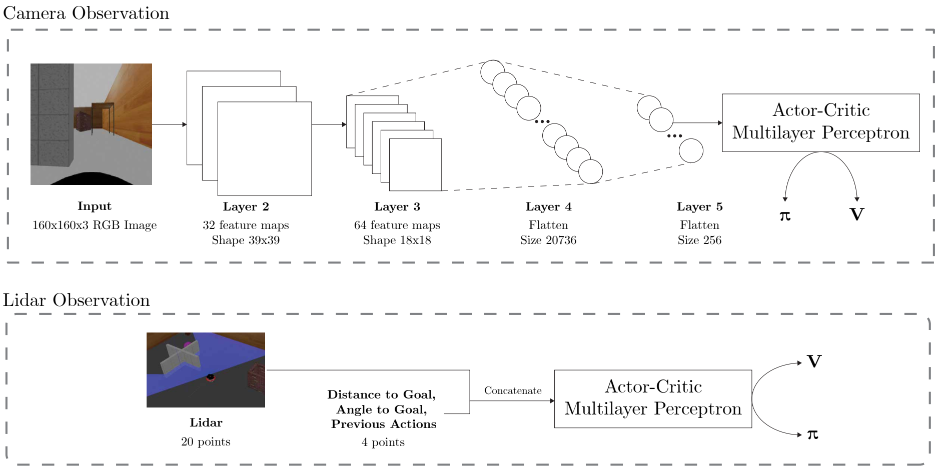


Fig. 4 Camera and Lidar observation networks. In camera observation models (TD3, PPO, PPO-LSTM) the image is first passed through convolutional layers and flattened to 256 features to reduce the dimensionality of the input. These features are then input to the actor-critic MLP. For the Lidar observation models (TD3, PPO), the input is already lowly dimensional and so the 20 Lidar readings are concatenated with the distance to goal, angle to goal, and previous actions and input to the MLP.

Table 1 TD3 training parameters (final values are bolded)

Parameter	Value
Learning Rate	0.03 / 0.003 / **0.0003** / 0.00003 (see LR study)
Tau	0.005
Learning Starts (after, steps)	5000
Batch Size	16384 (Lidar), 256 (Camera)
Discount Factor	0.99
Exploration Noise	0.3

Table 2 PPO and PPO LSTM training parameters (final values are bolded)

Parameter	Value
Learning Rate (Lidar)	0.03 / **0.003** / 0.0003 / 0.00003 (see LR study)
Learning Rate (Camera)	0.03 / 0.003 / **0.0003** / 0.00003 (see LR study)
Batch Size	256
Discount Factor	0.99

2.3 Experimental Scenario Settings

The experimental settings are shown in table 3. The Lidar observation space comprises 20 discrete points, taken at even intervals in a 180° arc around the robot. The camera image produces a 160x160 px RGB image and 64x64 px RGB image for DreamerV3 (the input to the DreamerV3 is required to be a power of two[2]). Lidar noise is Gaussian, and camera noise results in a complete blackout. The agent position input (for Lidar

[2] https://github.com/danijar/dreamerv3/issues/12

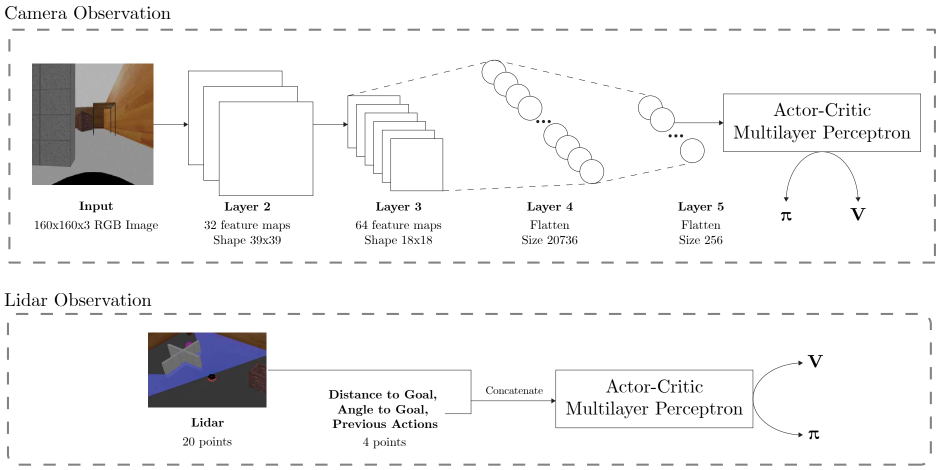


Fig. 4 相机和激光雷达观测网络。在相机观察模型（TD3、PPO、PPO-LSTM）中，图像首先通过卷积层并展平为 256 个特征，以减少输入的维度。然后将这些特征输入到 actor-critic MLP。对于激光雷达观测模型（TD3、PPO），输入已经是低维的，因此 20 个激光雷达读数与到目标的距离、到目标的角度以及之前的操作和 MLP 的输入连接在一起。

Table 1 TD3训练参数（最终值以粗体显示）

Parameter	Value
Learning Rate	0.03 / 0.003 / **0.0003** / 0.00003 (see LR study)
Tau	0.005
Learning Starts (after, steps)	5000
Batch Size	16384 (Lidar), 256 (Camera)
Discount Factor	0.99
Exploration Noise	0.3

Table 2 PPO和PPO LSTM训练参数（最终值以粗体显示）

Parameter	Value
Learning Rate (Lidar)	0.03 / **0.003** / 0.0003 / 0.00003 (see LR study)
Learning Rate (Camera)	0.03 / 0.003 / **0.0003** / 0.00003 (see LR study)
Batch Size	256
Discount Factor	0.99

2.3 Experimental Scenario Settings

实验设置如表 3 所示。激光雷达观测空间由 20 个离散点组成，这些点在机器人周围 180° 圆弧上以均匀间隔采集。相机图像为 DreamerV3 生成 160x160 px RGB 图像和 64x64 px RGB 图像（DreamerV3 的输入要求为 2[2] 的幂）。激光雷达噪声是高斯噪声，而相机噪声会导致完全停电。代理位置输入（对于激光雷达

[2]https://github.com/danijar/dreamerv3/issues/12

modality only) is fully accurate. The goal position for Lidar is random, and for the camera it is static. The time delta between steps is 0.05 seconds - this is the delay between which the agent makes actions. Note that the simulation runs at 4.0 times real-time speed, which results in this time step being 0.2 seconds, i.e. the agent runs at effectively 5 Hz in real time. The reward function produces a sparse reward when the agent reaches the goal, a penalty for colliding with a wall, and a small penalty each step to prevent inaction. The maximum steps per episode is 50.

There exist several random parameters within the simulation that affect the learning process. Randomness plays a key role, as shown by Mirowski et al. [6]. In their simulation, the RL agents outperformed humans on static small mazes but struggled on large complex mazes (with random parameters such as random agent spawning). Another difference between *DRL-Robot-Navigation* and DeepMind Lab is that our environment penalizes collisions with the wall, preventing wall-following strategies.

Camera policies require longer training compared with Lidar (greater than 500k steps) which exceeds the computational budget for this study. To simplify it, the goal is spawned at the centre of the map, instead of being spawned randomly. The robot is still spawned with a random position and orientation. This is akin to the 'static maze' from DeepMind Lab, as tested by Mirowski et al. Their 'random goal maze' is akin to our random goal position that we test during our Lidar experiments (albeit DeepMind Lab did not penalize wall collisions).

Generally, apart from DreamerV3, the camera policies likely take longer to converge to a solution due to increased observation space: even after convolutional layers, the state contains 256 features compared with the much simpler 24 points for Lidar (including previous actions, distance to goal, and angle to goal).

The original implementation of *DRL-Robot-Navigation* environment randomizes the starting position and orientation of the robot, as well as the position of the goal at the start of each episode.

To improve the convergence of camera policies, in order to better understand the sensor denial effects, we simplify the environment for our camera experiments. We fix the position of the goal to always be in the centre of the maze. This allows for less stochasticity in the environment and enables for quick convergence of the policy. For the Lidar experiments, we keep the spawn position of the goal and the robot random.

2.4 Training and Evaluation

The reinforcement learning (RL) models described in section 2.2 are trained with the following configurations and environments:

- Lidar (this includes distance and angle to goal). The models are trained on variety of 0x0 (no noise) maps, as well as maps with varying amounts of noise: 3x3, 5x5, and 7x7 metre noise zone.
- Camera only, static goal. The models are trained on variety of 0x0 (no noise) maps, as well as maps with varying amounts of noise: 3x3, 5x5, and 7x7 metre noise zone.

Unless otherwise stated, camera configurations contain a static goal to enable better convergence, as described in section 2.3. Each model is then evaluated on maps specified in the section 2.4, but generally this consists of being tested on each of the

仅模态）是完全准确的。激光雷达的目标位置是随机的，而相机的目标位置是静态的。步骤之间的时间增量为 0.05 秒 - 这是代理执行操作之间的延迟。请注意，模拟以 4.0 倍实时速度运行，这导致该时间步长为 0.2 秒，即代理实际上以 5 Hz 的速度实时运行。当智能体达到目标时，奖励函数会产生稀疏奖励、撞墙时的惩罚以及每一步的小惩罚以防止不作为。每集的最大步数为 50。

模拟中存在几个影响学习过程的随机参数。正如 Mirowski 等人所表明的那样，随机性起着关键作用。 [6]。在他们的模拟中，强化学习智能体在静态小型迷宫中的表现优于人类，但在大型复杂迷宫（具有随机参数，例如随机智能体生成）中表现不佳。 $DRL\text{-}Robot\text{-}Navigation$ 和 DeepMind Lab 之间的另一个区别是，我们的环境会惩罚与墙壁的碰撞，从而阻止沿墙策略。

与激光雷达相比，相机策略需要更长的训练时间（超过 50 万步），这超出了本研究的计算预算。为了简化它，目标是在地图中心生成的，而不是随机生成的。机器人仍然以随机位置和方向生成。这类似于 DeepMind 实验室的“静态迷宫”，由 Mirowski 等人测试。他们的“随机目标迷宫”类似于我们在激光雷达实验中测试的随机目标位置（尽管 DeepMind 实验室没有惩罚墙壁碰撞）。

一般来说，除了 DreamerV3 之外，由于观察空间增加，相机策略可能需要更长的时间才能收敛到解决方案：即使在卷积层之后，状态也包含 256 个特征，而激光雷达则包含简单得多的 24 个点（包括之前的动作、到目标的距离和到目标的角度）。

$DRL\text{-}Robot\text{-}Navigation$ 环境的原始实现随机化机器人的起始位置和方向，以及每集开始时的目标位置。

为了提高相机策略的收敛性，为了更好地理解传感器拒绝效应，我们简化了相机实验的环境。我们将目标的位置固定为始终位于迷宫的中心。这可以减少环境中的随机性，并实现策略的快速收敛。对于激光雷达实验，我们保持目标和机器人的生成位置是随机的。

2.4 Training and Evaluation

2.2 节中描述的强化学习 (RL) 模型使用以下配置和环境进行训练：

- 激光雷达（这包括到目标的距离和角度）。这些模型在各种 0x0（无噪声）地图以及具有不同噪声量的地图上进行训练：3x3、5x5 和 7x7 米噪声区域。
- 仅相机，静态目标。这些模型在各种 0x0（无噪声）地图以及具有不同噪声量的地图上进行训练：3x3、5x5 和 7x7 米噪声区域。

除非另有说明，相机配置包含一个静态目标，以实现更好的收敛，如第 2.3 节所述。然后在 2.4 节中指定的地图上评估每个模型，但通常这包括在每个模型上进行测试

Table 3 Simulation Parameters

	Parameter details
Lidar Measurements	180° LiDAR data discretized into 20 intervals
Camera Measurements	150px x 150px RGB image 64px x 64px RGB image (Dreamer)
Attack Type Lidar	Gaussian noise added to each discrete point
Attack Type Camera	Complete failure
Agent Position	Fully accurate (ground truth from Simulation)
Goal Position	Lidar: random Camera: static
Delay between steps	0.05 seconds
Reward function	$R(s,a) = \begin{cases} 1, & \text{goal reached} \\ -1, & \text{collision} \\ -1/50, & \text{otherwise} \end{cases} \quad (1)$
Max. Episode Steps	50

environments of differing noise size: 0x0, 3x3, 5x5, 7x7. In some cases, the model is evaluated on a simplified version of the map, as described in section 2.4.

Fig. 5 shows the outline of the training and evaluation setup for Lidar and camera environments.

2.4.1 Camera Evaluation Maps

For evaluation of camera policies, two different maps are used:

- *DRL-Robot-Navigation* map (same as training).
- A simplified *DRL-Robot-Navigation* map no obstacles, shown in Fig. 6. The goal is always static in the centre. The noise area is static in the centre of the map.

The purpose of using the simplified map is to better understand the behaviour of the policies. As the obstacles are taken out, the only task left is to get through the sensor denied area to the goal. As the noise is static, it always obstructs the goal. In this evaluation task we are ignoring the ability of the agent to avoid obstacles, but are focusing on whether the agent has learned to get through the noise area. The simplified map is only used for evaluation and is never used for training of the policies. By default, the *DRL-Robot-Navigation* is used for evaluation. The use of the simplified maps is explicitly specified in the results section.

3 Results

Several experiments are conducted to benchmark the different DRL algorithms on the environment and understand how different training regimes (vanilla vs adversarial)

Table 3 仿真参数

	Parameter details
Lidar Measurements	180° LiDAR data discretized into 20 intervals
Camera Measurements	150px x 150px RGB image 64px x 64px RGB image (Dreamer)
Attack Type Lidar	Gaussian noise added to each discrete point
Attack Type Camera	Complete failure
Agent Position	Fully accurate (ground truth from Simulation)
Goal Position	Lidar: random Camera: static
Delay between steps	0.05 seconds
Reward function	$$R(s,a) = \begin{cases} 1, & \text{goal reached} \\ -1, & \text{collision} \\ -1/50, & \text{otherwise} \end{cases} \quad (1)$$
Max. Episode Steps	50

不同噪声大小的环境：0x0、3x3、5x5、7x7。在某些情况下，模型在地图的简化版本上进行评估，如第 2.4 节所述。

图 5 显示了激光雷达和相机环境的训练和评估设置的概要。

2.4.1 Camera Evaluation Maps

为了评估相机策略，使用两个不同的地图：

- *DRL-Robot-Navigation* 地图（与训练相同）。
- 简化的 *DRL-Robot-Navigation* 地图没有障碍物，如图 6 所示。目标始终静止在中心。噪声区域在地图的中心是静态的。

使用简化地图的目的是为了更好地理解策略的行为。当障碍物被清除后，剩下的唯一任务就是穿过传感器禁区到达目标。由于噪音是静态的，它总是会阻碍目标。在这个评估任务中，我们忽略了智能体避开障碍物的能力，而是关注智能体是否学会了穿过噪声区域。简化地图仅用于评估，从不用于策略训练。默认情况下，使用 *DRL-Robot-Navigation* 进行评估。简化地图的使用在结果部分中明确指定。

3 Results

进行了多项实验来对环境中不同的 DRL 算法进行基准测试，并了解不同的训练方案（普通与对抗）如何

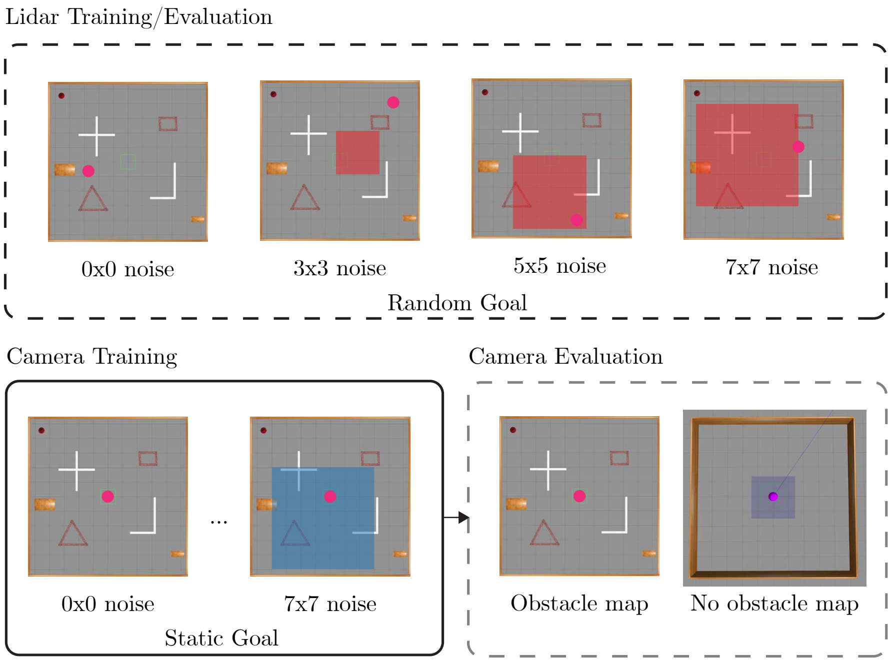


Fig. 5 Training and evaluation scenarios for Lidar and camera sensors. For Lidar scenarios, the models are trained on a variety of 0x0 (vanilla) maps and maps with varying amounts of sensor perturbation: 3x3, 5x5, and 7x7 metre sensor noise zones. These zones contain Gaussian noise which is added to the sensor readings. These trained policies are then evaluated on the same map. Likewise, for camera scenarios the models are trained on a variety of 0x0 (vanilla) maps and maps with varying amounts of sensor perturbation: 3x3, 5x5, and 7x7 metre sensor denial zones. Inside these zones, the camera sensor completely fails (all pixels turn to black). The models are trained on the default map, with a static goal in the middle, and the blue areas in which the camera completely fails. These policies are then evaluated on the default map with varying degree of noise. A second evaluation is performed on a map with no obstacles, to better understand the learned policies in sensor denied areas.

affect the performance of the models in environments with varying environmental perturbation size. While initially training of the Lidar and camera modalities is compared in section 3.1, the rest of the results section is split into two parts, based on modalities. The first part, in section 3.2 follows how PPO and TD3 perform on in terms of quantitative results, and then qualitatively in section 3.3 by understanding the paths followed by the policies, and their differences under different training regimes.

The camera observation section begins with a direct training comparison of the different algorithms on the vanilla environment in section 3.4, continues with a quantitative benchmark of the different models with sensor denial in section 3.5, and the models are evaluated on the no obstacle map in section 3.6. Finally, the results are discussed with respect to literature in section 3.7.

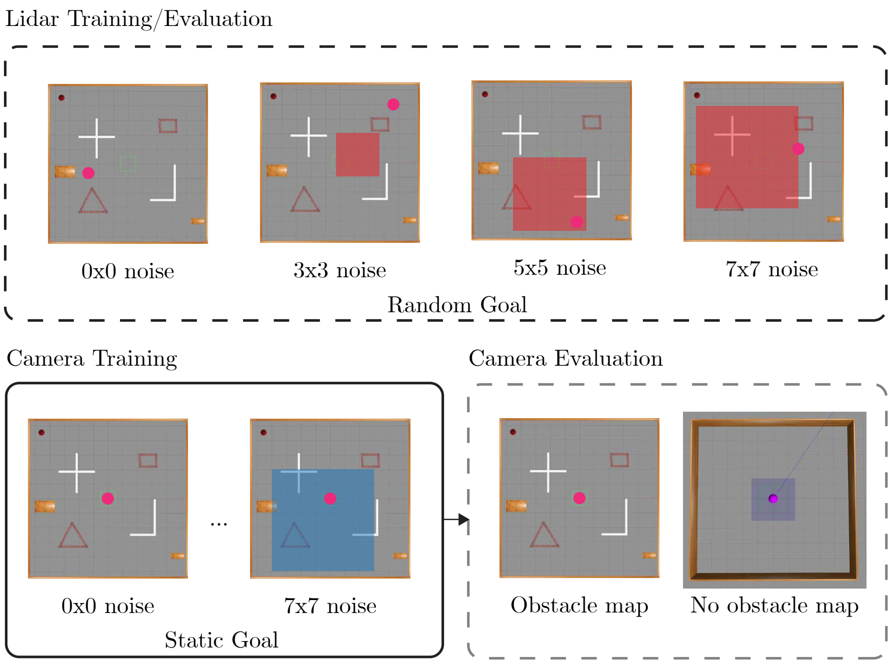
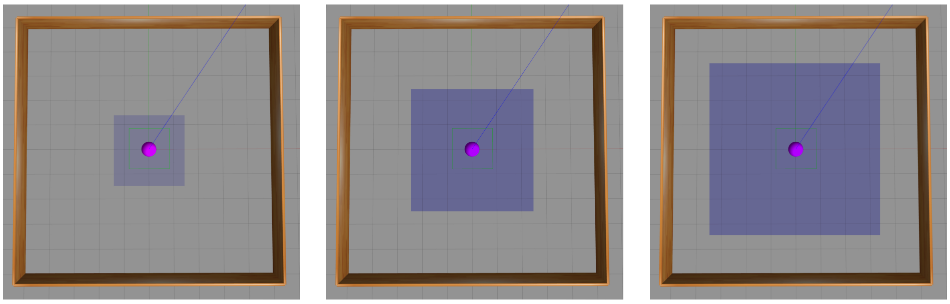


Fig. 5 激光雷达和摄像头传感器的训练和评估场景。对于激光雷达场景，模型在各种 0x0（普通）地图和具有不同传感器扰动量的地图上进行训练：3x3、5x5 和 7x7 米传感器噪声区域。这些区域包含添加到传感器读数中的高斯噪声。然后在同一张地图上评估这些经过训练的策略。同样，对于相机场景，模型在各种 0x0（普通）地图和具有不同传感器扰动量的地图上进行训练：3x3、5x5 和 7x7 米传感器拒绝区域。在这些区域内，相机传感器完全失效（所有像素变成黑色）。模型在默认地图上进行训练，中间有一个静态目标，蓝色区域是相机完全失效的区域。然后在具有不同程度噪声的默认地图上评估这些策略。第二次评估是在没有障碍物的地图上进行的，以更好地理解传感器拒绝区域中学到的策略。

影响模型在不同环境扰动大小的环境中的性能。虽然第 3.1 节对激光雷达和相机模式的初始训练进行了比较，但结果部分的其余部分根据模式分为两部分。第一部分在第 3.2 节中介绍了 PPO 和 TD3 在定量结果方面的表现，然后在第 3.3 节中通过了解政策遵循的路径以及它们在不同培训制度下的差异来定性。

摄像机观察部分首先在第 3.4 节中对普通环境下的不同算法进行直接训练比较，然后在第 3.5 节中对具有传感器拒绝的不同模型进行定量基准，并在第 3.6 节中在无障碍地图上对模型进行评估。最后，第 3.7 节中的文献讨论了结果。



(a) (b) (c)

Fig. 6 Simplified version of the *DRL-Robot-Navigation* map with no obstacles, with a static goal in the centre and a sensor perturbation area surrounding it. **(a)** shows the 3x3 sensor denied area, **(b)** shows the 5x5 sensor denied area, and **(c)** the 7x7 sensor denied area. The robot is spawned at a random position and orientation around the map. This map is only used to evaluate the camera policies, and not Lidar policies.

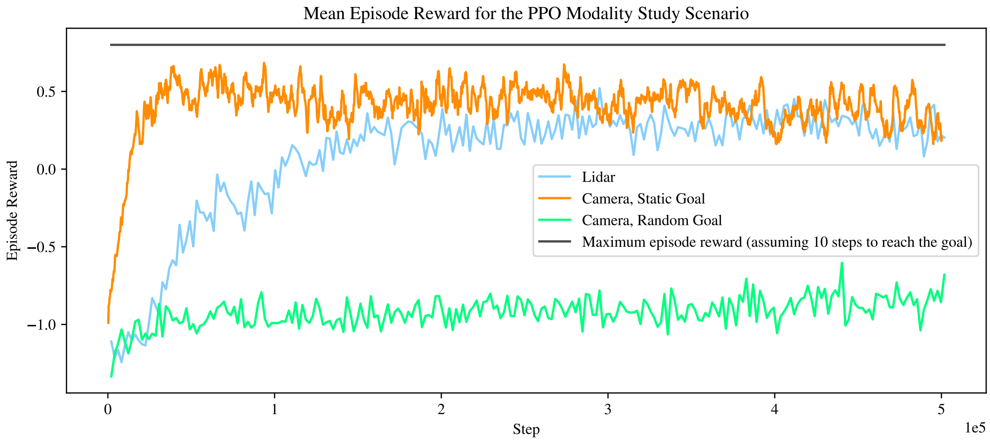
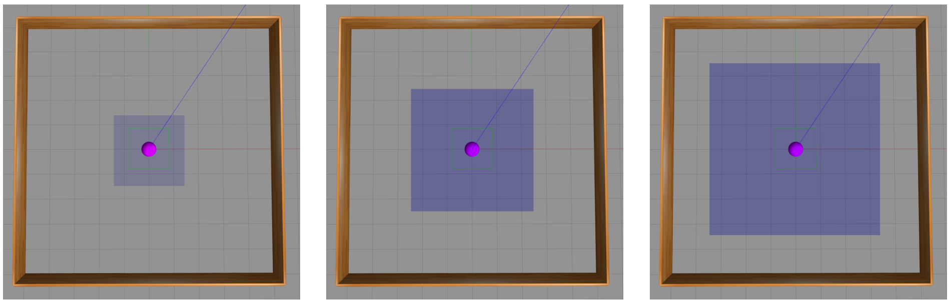


Fig. 7 Modality study using PPO for Lidar and camera. The camera was trained on static and random goals. Each of the models was trained for 500,000 steps.

3.1 Modality Study

The modality study shown in Fig. 7 shows the differences in model performance for the PPO algorithm across Lidar (with a random goal), camera with a static goal, and camera with a random goal. Lidar and camera with a static goal are quick to converge to a solution close to the maximum episode reward. Camera with a random goal does not converge to a solution in the given amount of steps. As we attempt to understand the effects of sensor perturbations, we will continue using camera with a static goal for the camera noise experiments.



(a) (b) (c)

Fig. 6 *DRL-Robot-Navigation* 地图的简化版本，没有障碍物，中心有一个静态目标，周围有一个传感器扰动区域。**(a)** 显示 3x3 传感器拒绝区域，**(b)** 显示 5x5 传感器拒绝区域，**(c)** 显示 7x7 传感器拒绝区域。机器人在地图周围的随机位置和方向生成。该地图仅用于评估相机策略，而不是激光雷达策略。

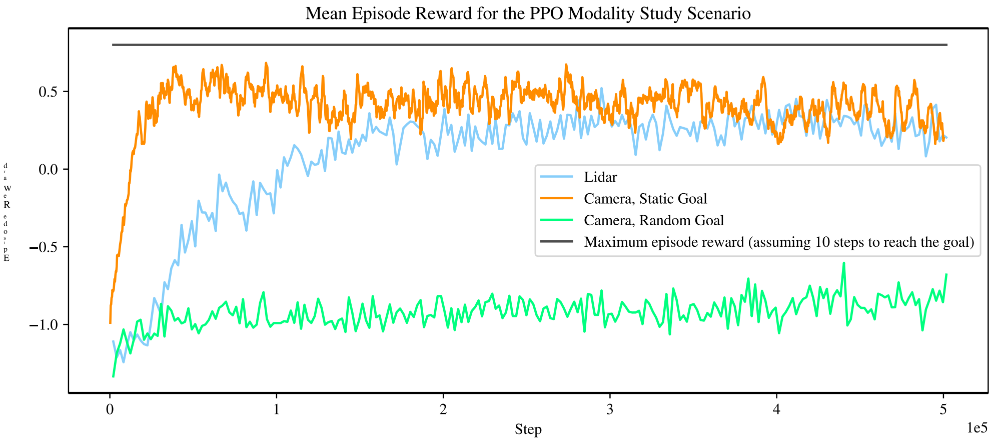


Fig. 7 使用 PPO 进行激光雷达和相机的模态研究。摄像机针对静态和随机目标进行训练。每个模型都经过 500,000 步的训练。

3.1 Modality Study

图 7 所示的模态研究显示了 PPO 算法在激光雷达（具有随机目标）、具有静态目标的相机和具有随机目标的相机上的模型性能差异。具有静态目标的激光雷达和摄像机可以快速收敛到接近最大剧集奖励的解决方案。具有随机目标的相机不会以给定的步数收敛到解决方案。当我们尝试了解传感器扰动的影响时，我们将继续使用具有静态目标的相机进行相机噪声实验。

3.2 Lidar Navigation with Noise

Although different algorithms are proposed in section 2.2, we were unable to find an appropriate learning rate to get PPO-LSTM to converge as shown in appendix A3. While it is possible to train DreamerV3 on the Lidar observation space, it is a large model for such a small observation space and we did not find that its performance increased over the other models. Hence, only TD3 and PPO are compared in this section.

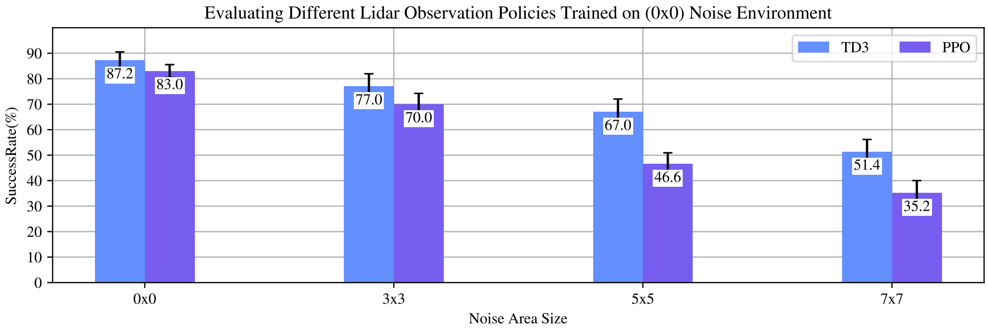


Fig. 8 Lidar noise area study for algorithms trained on the default environment with no noise. The algorithms were then evaluated on the default environment (with a random goal) with changing noise area sizes, as shown on the x-axis. The evaluation was performed for 100 episodes and repeated 5 times to establish a standard deviation.

Fig. 8 shows the evaluation results of the TD3, and PPO using the Lidar observation space on the default map. The algorithms are trained on the 0x0 (vanilla) map, and evaluated on environments of increasing Lidar noise area: 0x0 (vanilla), 3x3, 5x5, and 7x7 - referring to the size of the rectangle of the noise area in meters. TD3 outperforms PPO across the different evaluations. It achieves a mean success rate of 87.2% on the 0x0 evaluation scenario, compared with 83.0% for PPO. The performance differential between the two models generally increases with larger noise area sizes. When evaluated on the 7x7 noise area size, TD3 achieves 51.4%, while PPO achieves 35.2%.

Fig. 9 shows the same evaluation results of the TD3, and PPO, with the algorithms trained on the 3x3 noise map. In these results, the PPO closely matches the performance of TD3 across the evaluations, outperforming it on larger noise area sizes. When evaluated on the vanilla 0x0 scenario, TD3 achieves a mean success rate of 69.6% and PPO achieves 68.0%. This is a performance drop from the models trained on the vanilla 0x0 environments. When evaluated on the 7x7 scenario, the PPO model achieves a mean success rate of 54.0%. This is an improvement from the PPO model trained on the vanilla 0x0 environment and slightly outperforms the TD3 model trained on the same regime. The TD3 model trained on the 3x3 noise environment shows performance drops across all evaluation scenarios compared with the TD3 model trained on the 0x0 environment.

3.2 Lidar Navigation with Noise

尽管 2.2 节提出了不同的算法，但我们无法找到合适的学习率来使 PPO-LSTM 收敛，如附录 A3 所示。虽然可以在激光雷达观测空间上训练 DreamerV3，但对于如此小的观测空间来说，它是一个大型模型，我们没有发现它的性能比其他模型有所提高。因此，本节仅比较 TD3 和 PPO。

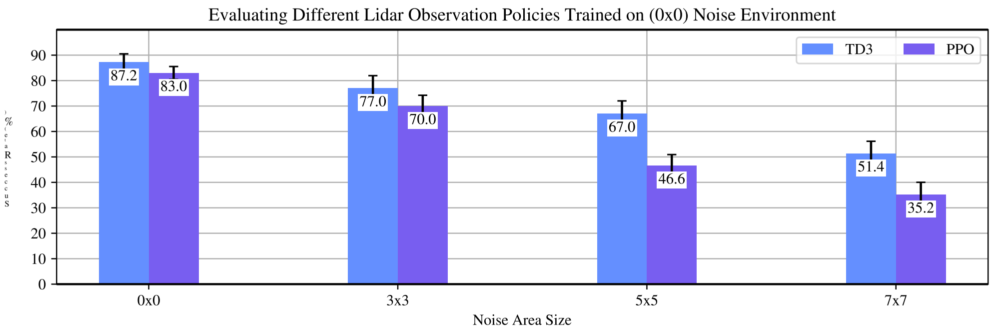


Fig. 8 激光雷达噪声区域研究，用于在无噪声的默认环境下训练的算法。然后在默认环境（具有随机目标）上评估算法，并改变噪声区域大小（如 x 轴所示）。评估进行了 100 次并重复 5 次以建立标准偏差。

图8显示了TD3和PPO在默认地图上使用激光雷达观测空间的评估结果。这些算法在 0x0（普通）地图上进行训练，并在激光雷达噪声区域增加的环境中进行评估：0x0（普通）、3x3、5x5 和 7x7 - 指的是噪声区域矩形的大小（以米为单位）。 TD3 在不同的评估中均优于 PPO。它在 0x0 评估场景上的平均成功率为 87.2%，而 PPO 的平均成功率为 83.0%。两个模型之间的性能差异通常随着噪声区域尺寸的增大而增大。当对 7x7 噪声区域大小进行评估时，TD3 达到 51.4%，而 PPO 达到 35.2%。

图 9 显示了 TD3 和 PPO 的相同评估结果，算法在 3x3 噪声图上进行训练。在这些结果中，PPO 在整个评估中与 TD3 的性能非常匹配，在较大的噪声区域尺寸上优于 TD3。在普通 0x0 场景中进行评估时，TD3 的平均成功率为 69.6%，PPO 的平均成功率为 68.0%。这是在普通 0x0 环境中训练的模型的性能下降。在 7x7 场景下进行评估时，PPO 模型的平均成功率为 54.0%。这是对在普通 0x0 环境上训练的 PPO 模型的改进，并且略优于在相同方案上训练的 TD3 模型。与在 0x0 环境中训练的 TD3 模型相比，在 3x3 噪声环境中训练的 TD3 模型在所有评估场景中的性能均有所下降。

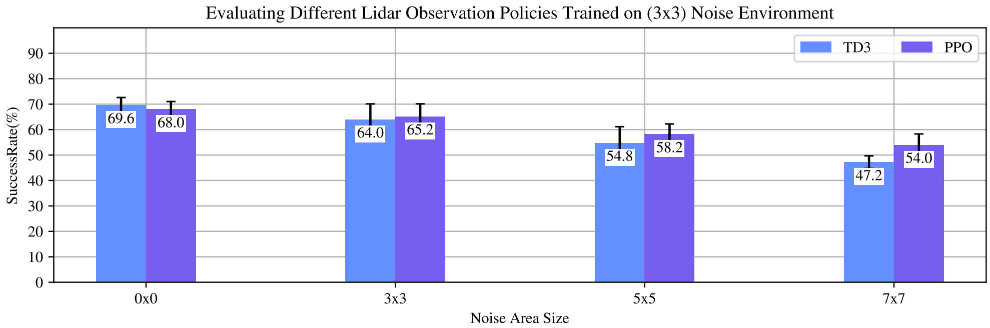


Fig. 9 Lidar noise area study for algorithms trained on the default environment with random 3x3 noise areas and random goal. The algorithms were then evaluated on the same map with different noise area sizes, as shown on the x-axis. The evaluation was performed for 100 episodes and repeated 5 times to establish a standard deviation.

3.3 Lidar Noise Training Regimes, Qualitative Results

As shown in section 3.2, the training regimes affect the evaluation performance of the algorithms. To better understand the behaviour of Lidar-specific policies, we can more closely investigate the effects of training in noisy regimes for each algorithm and investigate the trajectories of the robot during evaluation.

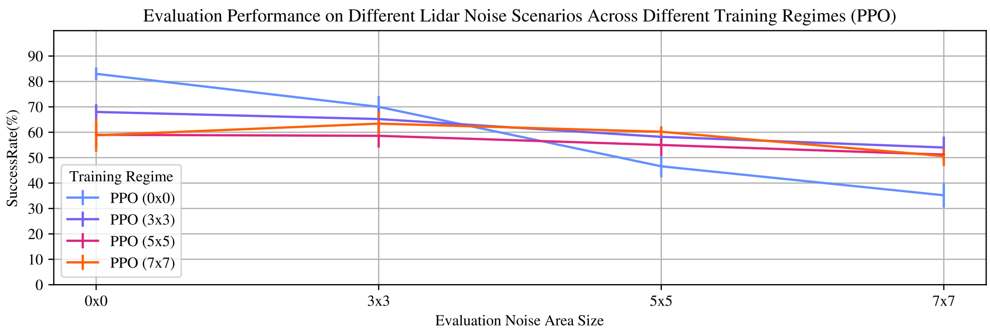


Fig. 10 PPO policies trained on different Lidar noise regimes evaluated on the default map (with random goal) with varying noise area sizes. The policies were evaluated for 100 steps 5 times, and the plot shows a mean and the standard error across those 5 repeated runs.

Fig. 10 shows the PPO policies trained on different Lidar noise regimes evaluated on the default map. It shows that the vanilla PPO (0x0) performs the best when evaluated on a map with no noise, with a mean success rate of 83%. The policies trained on noisy environments (PPO 3x3, 5x5, and 7x7) begin to outperform the vanilla policy when evaluated on the 5x5 noise map. While the noisy policies do not significantly degrade in performance as more noise is added to the environment, they show a consistent drop in performance compared with the vanilla model when evaluated on the 0x0 no

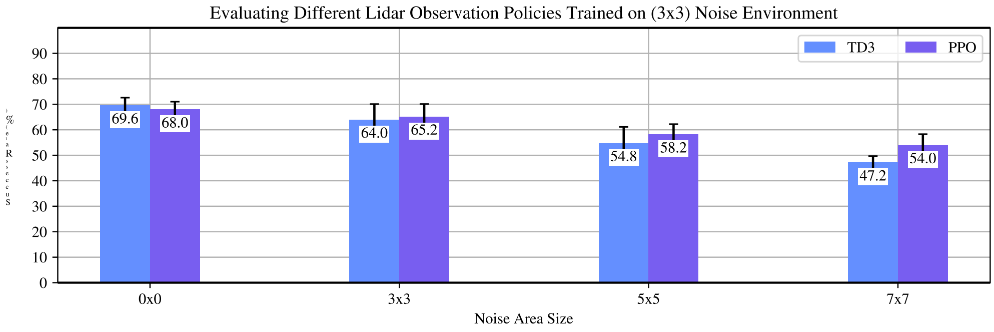


Fig. 9 激光雷达噪声区域研究，用于在具有随机 3x3 噪声区域和随机目标的默认环境下训练的算法。然后在具有不同噪声区域大小的同一张地图上评估算法，如 x 轴所示。评估进行了 100 次并重复 5 次以建立标准偏差。

3.3 Lidar Noise Training Regimes, Qualitative Results

如第 3.2 节所示，训练制度影响算法的评估性能。为了更好地理解激光雷达特定策略的行为，我们可以更仔细地研究每种算法在噪声环境中训练的效果，并在评估过程中研究机器人的轨迹。

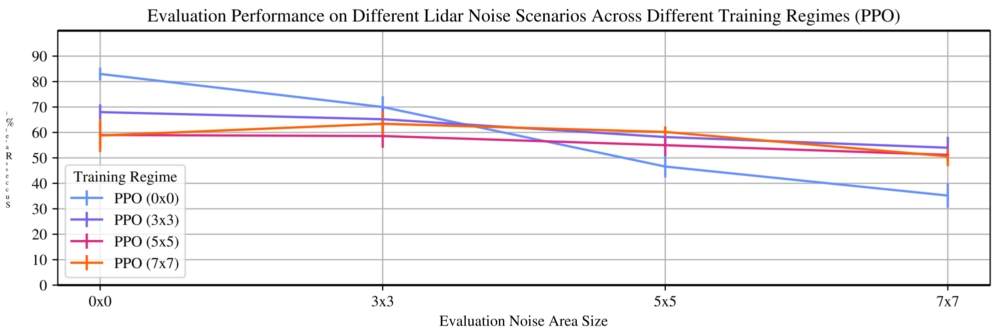


Fig. 10 在不同激光雷达噪声方案上训练的 PPO 策略在具有不同噪声区域大小的默认地图（具有随机目标）上进行评估。这些策略针对 100 个步骤进行了 5 次评估，该图显示了这 5 次重复运行的平均值和标准误差。

图 10 显示了在默认地图上评估的不同激光雷达噪声方案上训练的 PPO 策略。结果表明，在没有噪声的地图上进行评估时，普通 PPO (0x0) 表现最好，平均成功率为 83%。在 5x5 噪声图上进行评估时，在噪声环境（PPO 3x3、5x5 和 7x7）上训练的策略开始优于普通策略。虽然随着环境中添加更多噪声，噪声策略不会显着降低性能，但在 0x0 no 上进行评估时，与普通模型相比，它们的性能持续下降。

noise environment. This is likely due to the noisy policies following a riskier strategy, sacrificing collision avoidance capabilities to improve the navigation performance. To better understand this, we can qualitatively investigate the evaluation runs by plotting their paths, crashes and successful navigation.

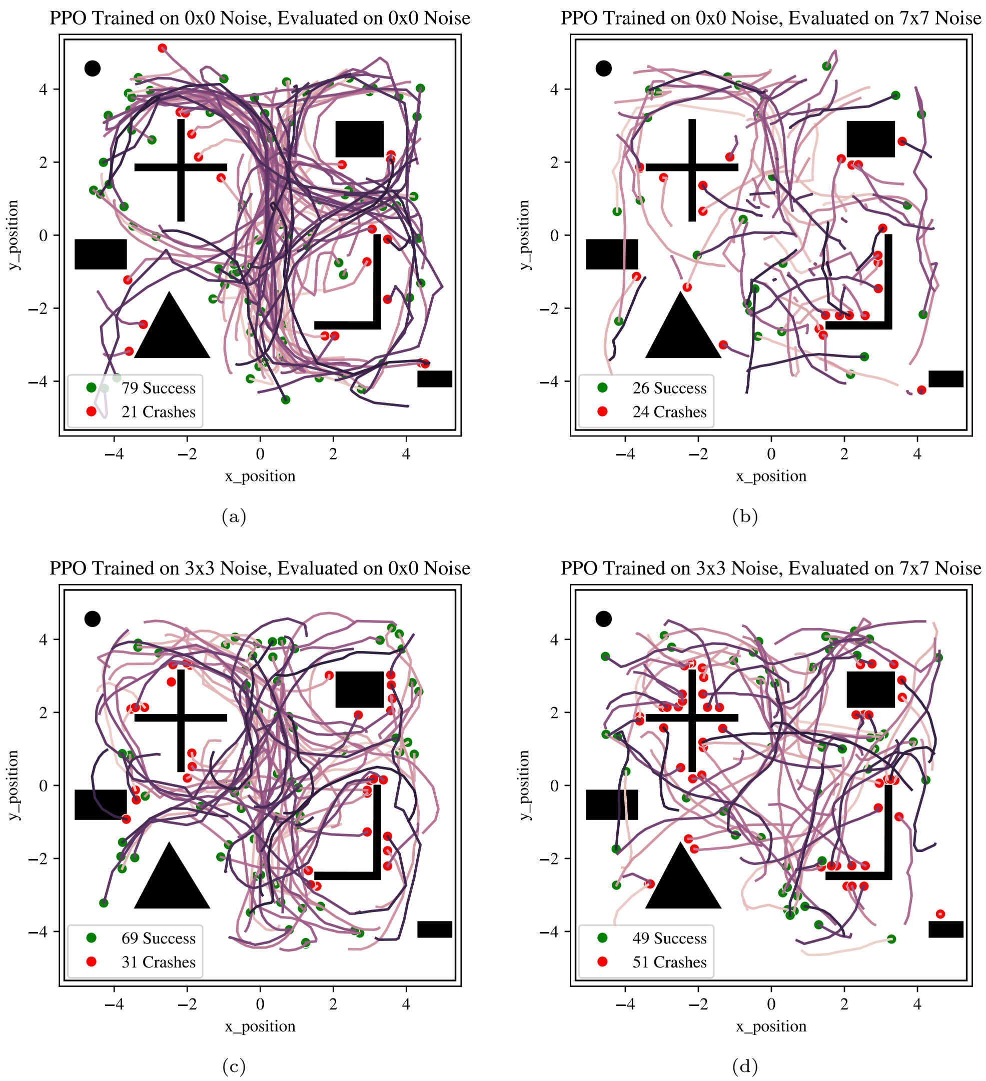


Fig. 11 100 path plots generated during evaluation by vanilla policy (PPO 0x0) evaluated on 0x0 noise map **(a)**, vanilla policy (PPO 0x0) evaluated on 7x7 noise map **(b)**, noise policy (PPO 3x3) evaluated on 0x0 noise map **(c)**, and noise policy (PPO 3x3) evaluated on 7x7 noise map **(d)**. The red dot shows a collision and the green dot shows a successful navigation to the goal.

噪声环境。这可能是由于遵循风险较高的策略的嘈杂策略，牺牲了防撞能力以提高导航性能。为了更好地理解这一点，我们可以通过绘制评估运行的路径、崩溃和成功导航来定性研究评估运行。

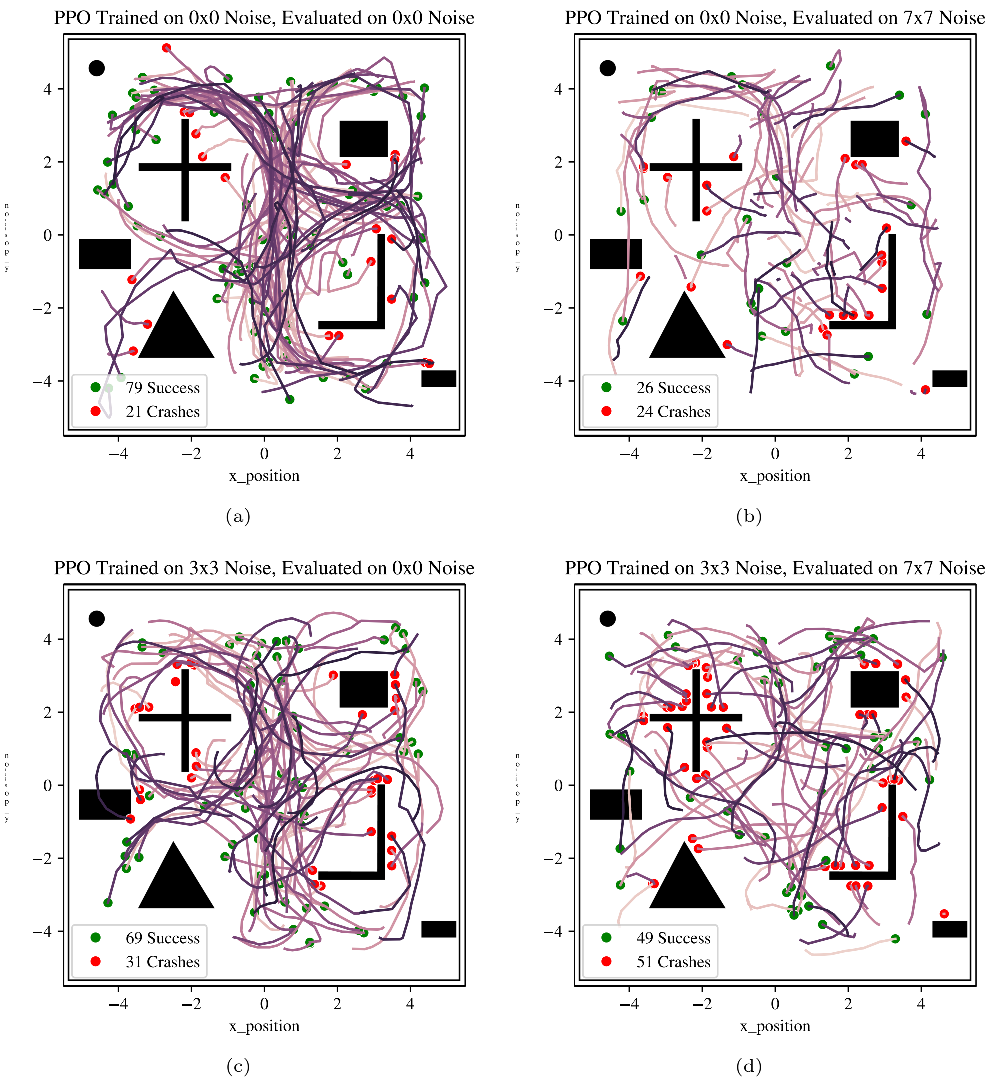


Fig. 11 评估过程中生成的 100 个路径图是通过在 0x0 噪声图 **(a)** 上评估的普通策略 (PPO 0x0)、在 7x7 噪声图 **(b)** 上评估的普通策略 (PPO 0x0)、在 0x0 噪声图 **(c)** 上评估的噪声策略 (PPO 3x3) 以及在 7x7 噪声图 **(d)** 上评估的噪声策略 (PPO 3x3) 进行评估。红点表示碰撞，绿点表示成功导航至目标。

Fig. 11 shows the 100 path plots generated during the evaluation of 2 different policies (PPO 0x0 and PPO 3x3) on two different evaluation maps (0x0 and 7x7). Fig. 11(a) shows the vanilla PPO evaluated on the no noise map. The same policy is evaluated on a map with random 7x7 noise areas in Fig. 11(b). The policy is unable to reach the targets and shows a significant decrease in performance (from 79 successes to 26 successes). It also shows that in half of the runs, the policy reached the maximum episode time step limit and it neither crashed nor reached the target. Compared with the evaluation on the no noise map, the paths tend to be much shorter, showing that the policy prefers to follow a 'safe' strategy of not doing anything when inside the noisy areas. In contrast, when trained on a little noise (3x3 random areas) and evaluated on no noise as shown in Fig. 11(c), the overall performance drops from 79 successful episodes to 69, and the number of crashes increases from 21 to 31. But, when evaluated on the 7x7 noise area as shown in Fig. 11(d), the policy no longer follows a 'safe' strategy and instead opts for a 'riskier' approach. This increases the number of successful episodes from 26 to 49 but also increases the amount of crashes from 24 to 51.

Overall, the safe policy did not result in better overall performance because the reward for reaching the max number of steps is -1, like with crashing, it is -1. But, because the risky policy results in more successful runs resulting in the positive +1 reward, it is the more optimal policy for the noisy evaluation maps. It is an interesting observation that the riskier strategies consistently emerged across the PPO runs when trained on noisy maps, while when not trained on any noise, the policy followed a safer strategy of letting the episodes run out when it was not certain of its sensor readings, as shown in Fig. 10. This is likely due to the noisy sensor readings falling outside of the previously observed domain for the policy trained on the non noisy environment.

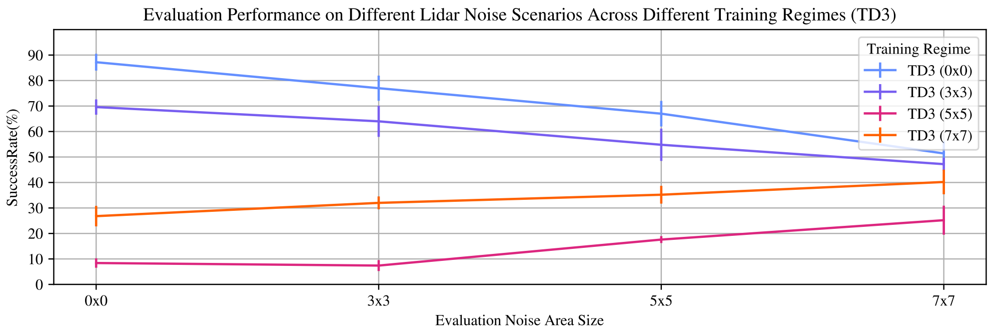


Fig. 12 TD3 policies trained on different Lidar noise regimes evaluated on the default map (with random goal) with varying noise area sizes. The policies were evaluated for 100 steps 5 times, and the plot shows a mean and the standard error across those 5 repeated runs.

Fig. 12 likewise shows the TD3 policies. Unlike in the PPO training schemes, TD3 generally did not benefit from being trained on noisy scenarios and the default policy trained on no-noise map performed the best across all evaluation scenarios.

图 11 显示了在两个不同的评估图（0x0 和 7x7）上评估 2 个不同策略（PPO 0x0 和 PPO 3x3）期间生成的 100 个路径图。图 11（a）显示了在无噪声图上评估的普通 PPO。在图 11(b) 中，在具有随机 7x7 噪声区域的地图上评估相同的策略。该政策无法达到目标，并且绩效显着下降（从 79 次成功降至 26 次成功）。它还表明，在一半的运行中，该策略达到了最大事件时间步长限制，并且既没有崩溃也没有达到目标。与无噪声地图上的评估相比，路径往往要短得多，这表明该策略更倾向于遵循“安全”策略，即在噪声区域内不做任何事情。相比之下，当在小噪声（3x3随机区域）上进行训练并在无噪声上进行评估时，如图11（c）所示，整体性能从79次成功下降到69次，崩溃次数从21次增加到31次。但是，当在7x7噪声区域上进行评估（如图11（d）所示）时，该策略不再遵循“安全”策略，而是选择“风险更高”的方法。这将成功事件的数量从 26 次增加到 49 次，但也将崩溃次数从 24 次增加到 51 次。

总的来说，安全策略并没有带来更好的整体性能，因为达到最大步数的奖励是 -1，就像崩溃一样，它是 -1。但是，由于有风险的策略会导致更成功的运行，从而产生正的 +1 奖励，因此对于噪声评估图而言，它是更优化的策略。一个有趣的观察结果是，当在噪声地图上进行训练时，在 PPO 运行中始终出现风险较高的策略，而当未对任何噪声进行训练时，该策略会遵循一种更安全的策略，即在不确定其传感器读数时让事件运行完，如图 10 所示。这可能是由于噪声传感器读数超出了在非噪声环境上训练的策略之前观察到的域。

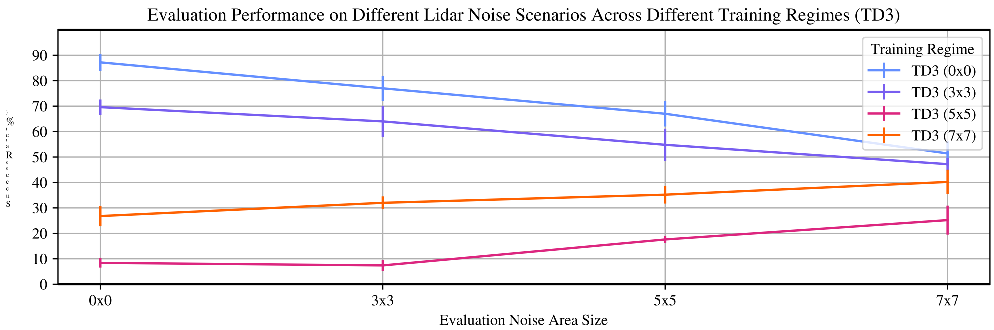


Fig. 12 TD3 策略在不同激光雷达噪声方案上进行训练，并在具有不同噪声区域大小的默认地图（具有随机目标）上进行评估。这些策略针对 100 个步骤进行了 5 次评估，该图显示了这 5 次重复运行的平均值和标准误差。

图12同样显示了TD3策略。与 PPO 训练方案不同，TD3 通常不会从噪声场景的训练中受益，并且在无噪声图上训练的默认策略在所有评估场景中表现最佳。

3.4 Algorithm Training Comparison (Camera Observation)

As described in section 2.2, different RL methods are trained on the *DRL-Robot-Navigation* environment: TD3, PPO, Recurrent PPO (or PPO-LSTM), and DreamerV3.

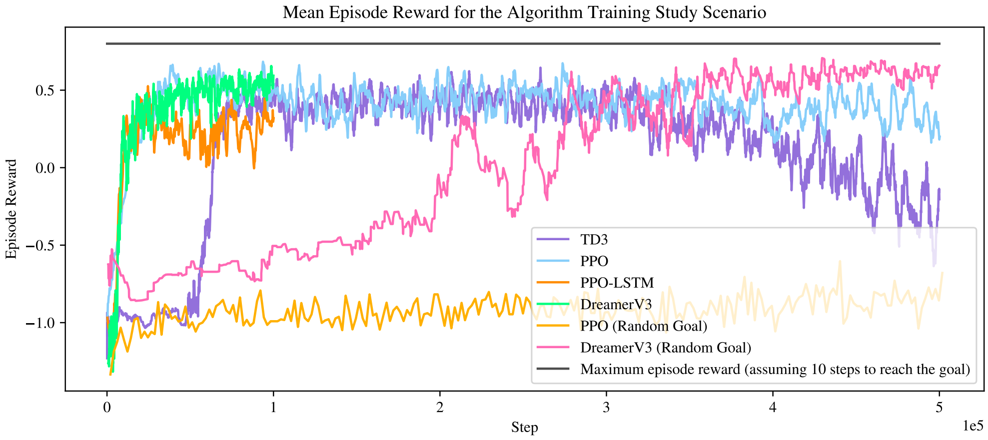


Fig. 13 Comparison of convergence of different algorithms during training. DreamerV3 was trained with a random goal, while the other models were trained using a static goal in the centre (making DreamerV3s convergence more complicated).

Fig. 13 shows the averaged episode reward during training for the different algorithms, across 500,000 steps of training. For the static goal scenario, TD3, PPO, and PPO-LSTM converge to a solution quickly, each one reaching 0.5 mean reward in less than 100,000 steps. The only model that was able to learn to navigate to a random goal was DreamerV3. By comparison, PPO failed to learn to navigate even after 500,000 steps. Nonetheless, to better understand the effects of noise on training, a static goal will be used for the camera policies.

3.5 Camera Navigation with Sensor Denial

Fig. 14 shows the evaluation results of the different algorithms, TD3, PPO, PPO-LSTM, and DreamerV3 using the camera observation space. The algorithms are trained on the vanilla 0x0 map, and evaluated on environments of increasing camera denial area: 0x0 (sensor denial), 3x3, 5x5, and 7x7 - referring to the size of the rectangle of the sensor denied area in meters. On the 0x0 evaluation scenario, DreamerV3 achieved the highest mean success rate of 90.8%, with TD3 achieving a similar performance of 88.8%. PPO-LSTM was third with a mean of 82.2% and PPO performed the worst with a mean success of 71.6%. On the 3x3 evaluation scenario, the pattern remains the same, with DreamerV3 achieving the highest mean of 73.2% (with a much larger standard error of 6.8), TD3 close behind with 68.4%, PPO-LSTM third

3.4 Algorithm Training Comparison (Camera Observation)

如 2.2 节所述，在 *DRL- Robot-Navigation* 环境上训练不同的 RL 方法：TD3、PPO、Recurrent PPO（或 PPO-LSTM）和 DreamerV3。

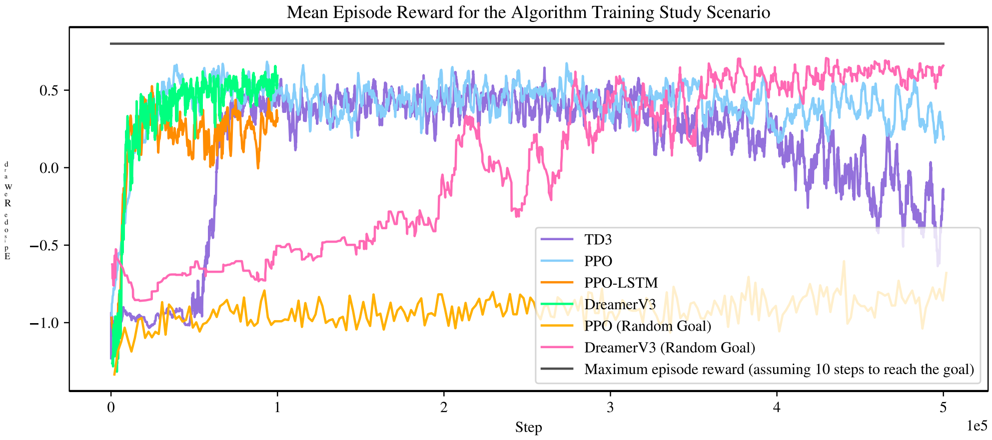


Fig. 13 训练过程中不同算法的收敛性比较。DreamerV3 使用随机目标进行训练，而其他模型则使用中心的静态目标进行训练（使得 DreamerV3 的收敛更加复杂）。

图 13 显示了不同算法训练期间 500,000 个训练步骤的平均情节奖励。对于静态目标场景，TD3、PPO 和 PPO-LSTM 快速收敛到一个解决方案，每一个都在不到 100,000 步的时间内达到 0.5 的平均奖励。唯一能够学习导航到随机目标的模型是 DreamerV3。相比之下，即使走了 50 万步，PPO 也无法学会导航。尽管如此，为了更好地了解噪声对训练的影响，相机策略将使用静态目标。

3.5 Camera Navigation with Sensor Denial

图14显示了使用相机观察空间的不同算法TD3、PPO、PPO-LSTM和DreamerV3的评估结果。这些算法在普通 0x0 地图上进行训练，并在增加相机拒绝区域的环境中进行评估：0x0（传感器拒绝）、3x3、5x5 和 7x7 - 指传感器拒绝区域的矩形大小（以米为单位）。在 0x0 评估场景中，DreamerV3 实现了最高的平均成功率 90.8%，TD3 实现了类似的性能 88.8%。PPO-LSTM 排名第三，平均成功率为 82.2%，PPO 表现最差，平均成功率为 71.6%。在 3x3 评估场景中，模式保持不变，DreamerV3 达到了最高平均值 73.2%（标准误差更大，为 6.8），TD3 紧随其后，为 68.4%，PPO-LSTM 第三

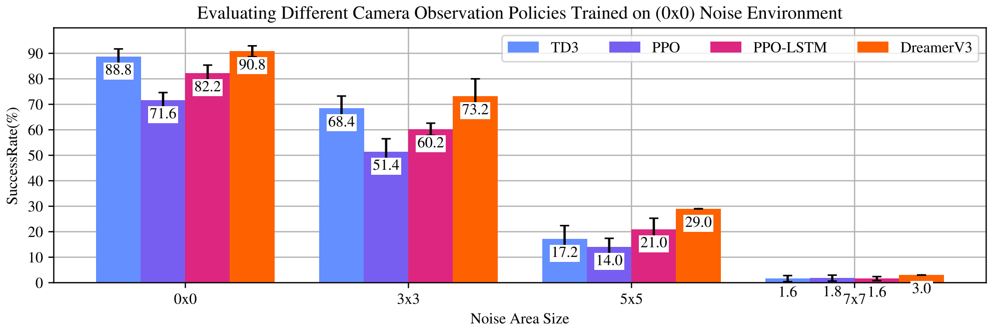


Fig. 14 Camera noise area study for algorithms trained on the vanilla environment with no sensor denial. The algorithms were then evaluated on the vanilla environment with changing noise area sizes, as shown on the x-axis. The evaluation was performed for 100 episodes and repeated 5 times to establish a standard deviation.

and PPO last. The evaluation scenarios for 5x5 and 7x7 show severe degradation in performance across all the algorithms.

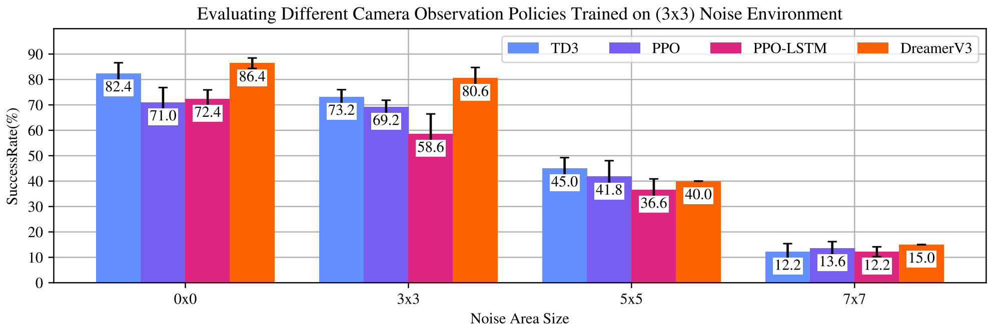


Fig. 15 Camera noise are study, for algorithms trained on the default environment with 3x3 sensor denial zones. The algorithms were then evaluated on the default environment with changing sensor denial area sizes, as shown on the x-axis. The evaluation was performed for 100 episodes and repeated 5 times to present a mean and standard error.

Similarly, Fig. 15 shows the evaluation results of the algorithms trained on a map with random 3x3 denied areas. When evaluated on the 0x0 scenario, a similar pattern emerges with DreamerV3 achieving the highest mean success rate of 86.4% (albeit, this is a performance drop compared with 90.8% when trained on the clean scenario). TD3 is second with 82.4%, and PPO and PPO-LSTM show the worst performance achieving 71.0% and 72.4% respectively. All of the policies show performance drops compared to policies trained on the clean (i.e. no sensor denied areas) environment.

When evaluated on the 3x3 scenario, DreamerV3 performs the best achieving 80.6% mean success rate. This is an improvement from the vanilla model (from 73.2%). TD3 mean success rate is 73.2%, PPO 69.2%, and PPO-LSTM is last with 58.6%. Likewise,

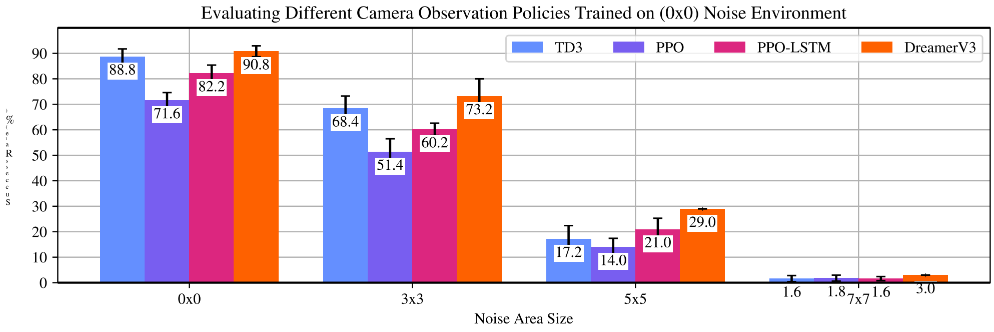


Fig. 14 相机噪声区域研究，用于在没有传感器拒绝的普通环境下训练的算法。然后在噪声区域大小不断变化的普通环境中评估算法，如 x 轴所示。评估进行了 100 次并重复 5 次以建立标准偏差。

最后是 PPO。 5x5 和 7x7 的评估场景显示所有算法的性能都严重下降。

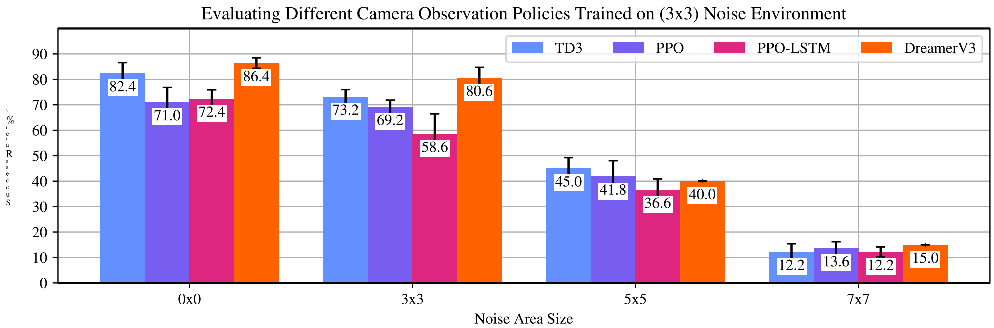


Fig. 15 研究了相机噪声，用于在具有 3x3 传感器拒绝区域的默认环境中训练的算法。然后在默认环境下评估算法，并改变传感器拒绝区域大小（如 x 轴所示）。评估进行了 100 次并重复 5 次以呈现平均值和标准误差。

类似地，图 15 显示了在具有随机 3x3 拒绝区域的地图上训练的算法的评估结果。在 0x0 场景中进行评估时，出现了类似的模式，DreamerV3 实现了 86.4% 的最高平均成功率（尽管与在干净场景中训练时的 90.8% 相比，性能有所下降）。 TD3 位居第二，为 82.4%，PPO 和 PPO-LSTM 表现最差，分别为 71.0% 和 72.4%。与在干净（即没有传感器拒绝区域）环境中训练的策略相比，所有策略的性能均有所下降。

在 3x3 场景中进行评估时，DreamerV3 表现最佳，平均成功率达到 80.6%。这是对普通模型（73.2%）的改进。 TD3 平均成功率为 73.2%，PPO 为 69.2%，PPO-LSTM 垫底，为 58.6%。同样地，

TD3 and PPO also show improvements from the vanilla models. PPO-LSTM is the only model that did not report an improvement from the vanilla model.

All models report an improvement in evaluation performance on 5x5 and 7x7 noise area maps (compared with models trained on no noise in the environment), showing that training on some noise in the scenario generally improves the evaluation performance on noisy evaluation scenarios. Still, these improvements are not so significant that the models are reliably achieving the goal. The downside of training the model on the 3x3 scenario with sensor-denied areas is that these models appear to suffer from performance degradation when evaluated on the 0x0 scenario.

3.6 Camera Navigation with Sensor Denial (No Obstacle Map)

Section 3.5 shows that the training regimes have an effect on policy behaviour in sensor-denied areas. To better understand this behaviour, they are evaluated on the simplified, no obstacle map, as described in section 2.4.

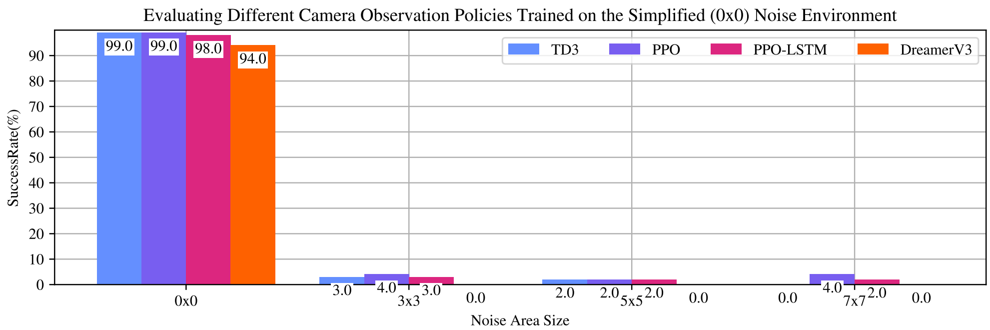


Fig. 16 Camera noise area study for algorithms trained on the default environment with 0x0 noise. The algorithms were then evaluated on the simplified map with changing noise area sizes, as shown on the x-axis. The evaluation was performed for 100 episodes.

Fig. 16 shows the evaluation results of the different algorithms, TD3, PPO, PPO-LSTM, and DreamerV3 using the camera observation space. The algorithms are trained on the vanilla 0x0 (no sensor denial) default map, and evaluated on the simplified, no obstacle map, with variations of increasing, static camera denial area which is located in the centre of the map: 0x0 (no sensor denial), 3x3, 5x5, and 7x7 - referring to the size of the rectangle of the sensor denied area in meters. All four of the models exhibit similar behaviour of successfully navigating in nearly all of the episodes on the 0x0 evaluation, but then consistently failing when any noise is introduced (with a few successes likely caused by spawning near the goal).

Similarly, Fig. 17 shows the evaluation results of the different algorithms trained on the default map with random 3x3 noise areas. The 0x0 evaluation does not significantly change, with TD3, PPO, and DreamerV3 still successfully navigating to the target 98, 98, and 96 times out of 100. For the 3x3 evaluation, TD3 reached the goal 74 times, PPO 71 times, PPO-LSTM 66 times, and DreamerV3 showed the best performance

TD3 和 PPO 也显示出相对于普通型号的改进。 PPO-LSTM 是唯一一个没有报告比普通模型有改进的模型。

所有模型都报告在 5x5 和 7x7 噪声区域图上的评估性能有所提高（与在环境中无噪声的情况下训练的模型相比），这表明对场景中的某些噪声进行训练通常会提高在有噪声的评估场景中的评估性能。尽管如此，这些改进并没有那么显着，以至于模型无法可靠地实现目标。在具有传感器拒绝区域的 3x3 场景中训练模型的缺点是，在 0x0 场景中进行评估时，这些模型似乎会出现性能下降的问题。

3.6 Camera Navigation with Sensor Denial (No Obstacle Map)

3.5 节表明，训练制度对传感器拒绝区域的政策行为有影响。为了更好地理解这种行为，我们在简化的无障碍地图上对它们进行评估，如第 2.4 节所述。

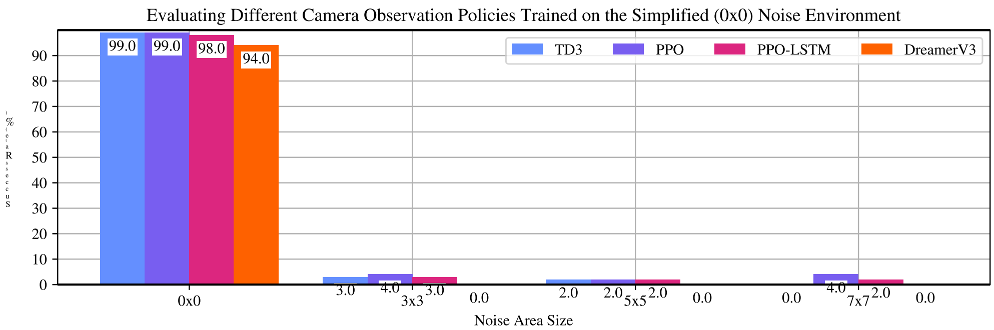


Fig. 16 相机噪声区域研究，用于在具有 0x0 噪声的默认环境下训练的算法。然后在具有变化的噪声区域大小的简化地图上评估算法，如 x 轴所示。对100集进行了评估。

图16显示了使用相机观察空间的不同算法TD3、PPO、PPO-LSTM和DreamerV3的评估结果。这些算法在普通 0x0（无传感器拒绝）默认地图上进行训练，并在简化的无障碍物地图上进行评估，其中位于地图中心的静态相机拒绝区域不断增加：0x0（无传感器拒绝）、3x3、5x5 和 7x7 - 指的是传感器拒绝区域的矩形大小（以米为单位）。所有四个模型都表现出类似的行为，即在 0x0 评估的几乎所有场景中都成功导航，但在引入任何噪声时始终失败（少数成功可能是由于在目标附近生成而导致的）。

同样，图 17 显示了在具有随机 3x3 噪声区域的默认地图上训练的不同算法的评估结果。 0x0 评估没有显着变化，TD3、PPO 和 DreamerV3 在 100 次中仍然成功导航到目标 98、98 和 96 次。对于 3x3 评估，TD3 达到目标 74 次，PPO 71 次，PPO-LSTM 66 次，DreamerV3 表现最好

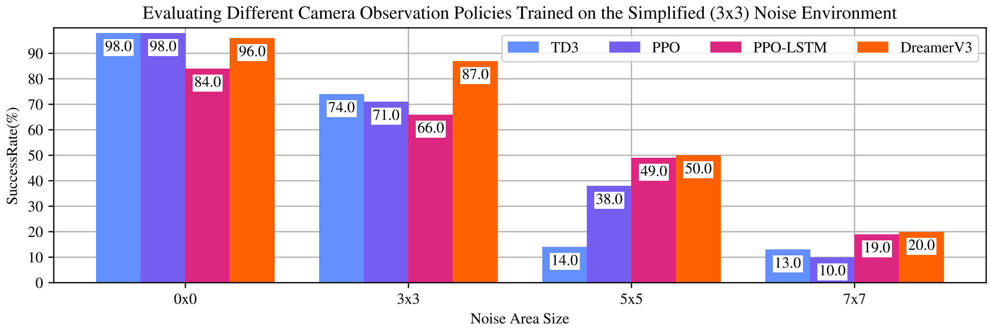


Fig. 17 Camera noise are study, for algorithms trained on the default environment with 3x3 noise. The algorithms were then evaluated on the simplified map with changing noise area sizes, as shown on the x-axis. The evaluation was performed for 100 episodes.

with 87 times. This is a significant improvement over the models trained on the 0x0 (no-noise) environment - a sign that the policies have learned to navigate to the goal with noise present. The 3x3-trained models also show performance improvements over the 0x0-trained models when evaluated on 5x5 and 7x7 scenarios. However, they do not achieve high (90+) success rates, which would be an indication of the policies truly solving the scenarios.

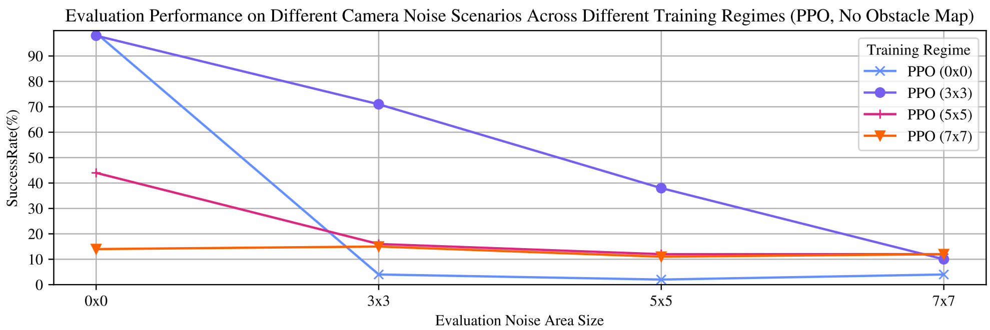


Fig. 18 PPO policies trained on different noise regimes evaluated on the simplified map with varying noise area sizes.

Fig. 18 shows the PPO policies trained on different camera noise regimes evaluated on the no-obstacle map. It shows that the PPO (0x0) and PPO (3x3) perform the best when evaluated on the 0x0 scenario, achieving 99 and 98 successful episodes. The PPO (0x0) fails to navigate to the goal when evaluated on any noise maps. PPO (5x5) and PPO (7x7) both failed to navigate successfully in the 0x0 evaluation, a sign that too much noise hindered the learning of those models.

Fig. 19 shows the 100 path plots generated during the evaluation of 2 different policies (PPO 0x0 and PPO 3x3) on two different evaluation maps (0x0 and 3x3 static

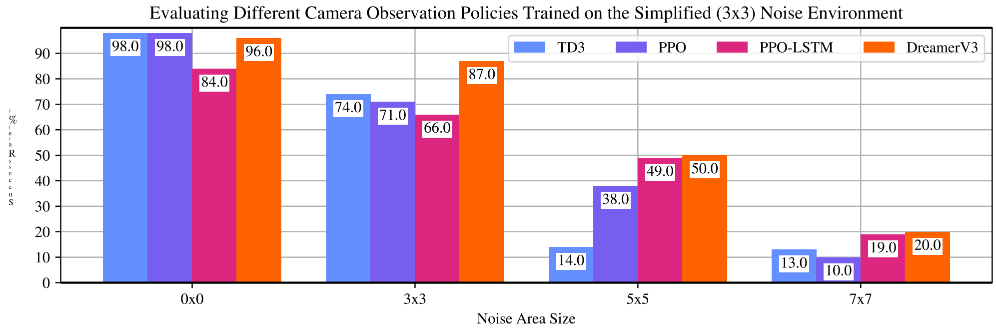


Fig. 17 研究相机噪声，用于在具有 3x3 噪声的默认环境下训练的算法。然后在具有变化的噪声区域大小的简化地图上评估算法，如 x 轴所示。对100集进行了评估。

与 87 次。与在 0x0（无噪声）环境中训练的模型相比，这是一个显着的改进 - 这表明策略已经学会在存在噪声的情况下导航到目标。在 5x5 和 7x7 场景中进行评估时，3x3 训练的模型还显示出比 0x0 训练的模型有性能提升。然而，它们并没有达到很高的成功率 (90+)，这表明政策真正解决了这些场景。

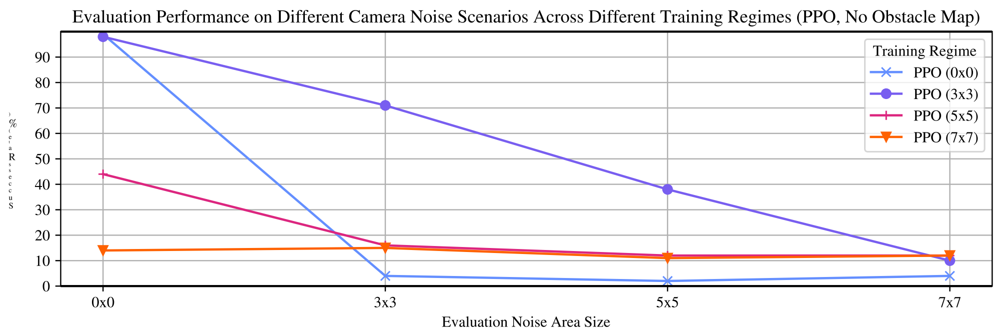


Fig. 18 针对不同噪声状况进行训练的 PPO 政策在具有不同噪声区域大小的简化地图上进行评估。

图 18 显示了在无障碍地图上评估的不同相机噪声方案上训练的 PPO 策略。结果表明，在 0x0 场景下评估时，PPO (0x0) 和 PPO (3x3) 表现最好，分别实现了 99 次和 98 次成功。在任何噪声图上进行评估时，PPO (0x0) 无法导航到目标。 PPO (5x5) 和 PPO (7x7) 在 0x0 评估中均未能成功导航，这表明过多的噪声阻碍了这些模型的学习。

图 19 显示了在两个不同的评估图（0x0 和 3x3 静态）上评估 2 个不同策略（PPO 0x0 和 PPO 3x3）期间生成的 100 个路径图。

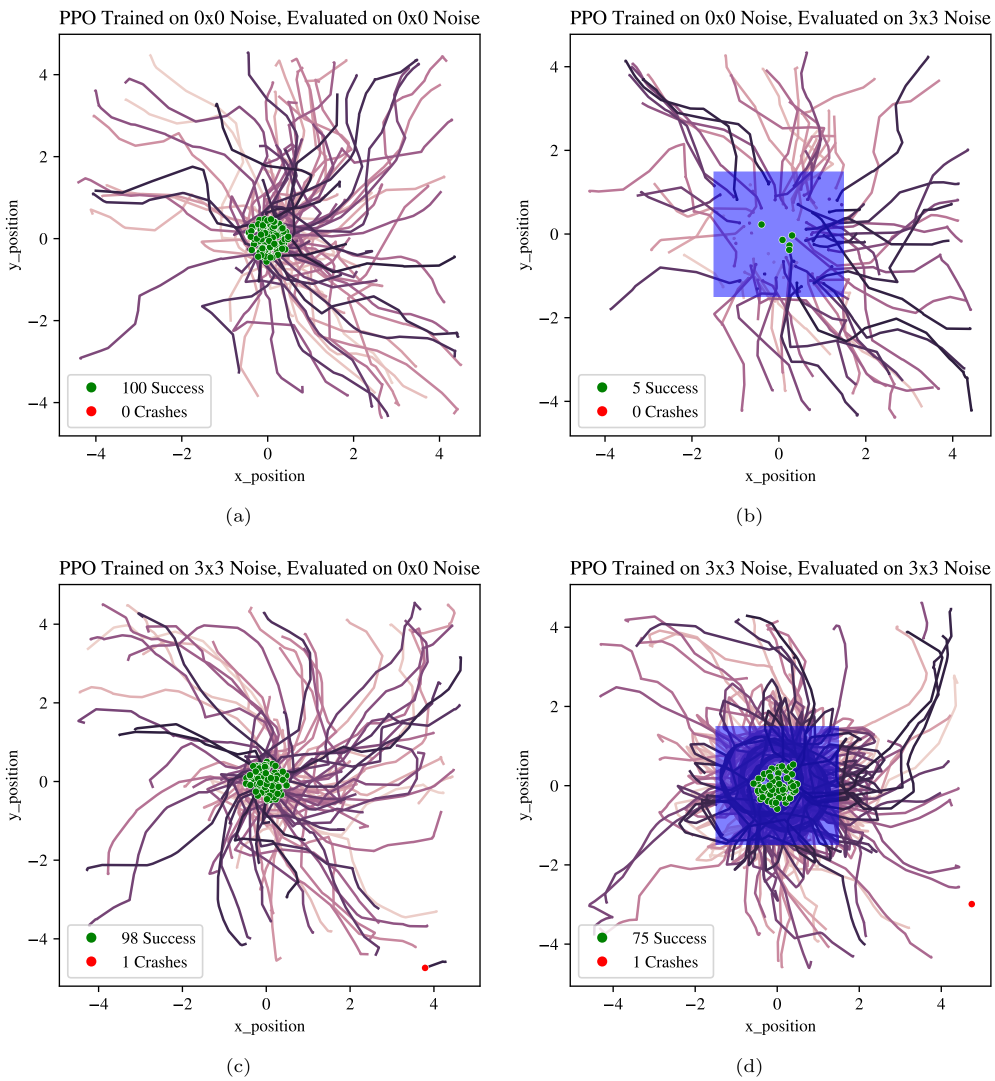


Fig. 19 100 path plots generated during evaluation by vanilla policy (PPO 0x0) evaluated on 0x0 map **(a)**, vanilla policy (PPO 0x0) evaluated on 3x3 sensor denial map **(b)**, adversarially trained (PPO 3x3) evaluated on 0x0 sensor denial map **(c)**, and adversarially trained (PPO 3x3) evaluated on 3x3 sensor denial map **(d)**. Red dot shows a collision, green dot shows a successful navigation to the goal. Blue zone shows the area of sensor denial.

sensor denial zones, static goal in the centre, and no obstacles present). Fig. 19(a) shows the vanilla PPO evaluated on the 0x0 map. The same vanilla policy is evaluated on a map with a static 3x3 noise area around the goal in Fig. 19(b). The policy is unable to navigate to the goal and shows a significant decrease in performance (from 100 successes to 5 successes). When the robot enters the noise area, the camera fails,

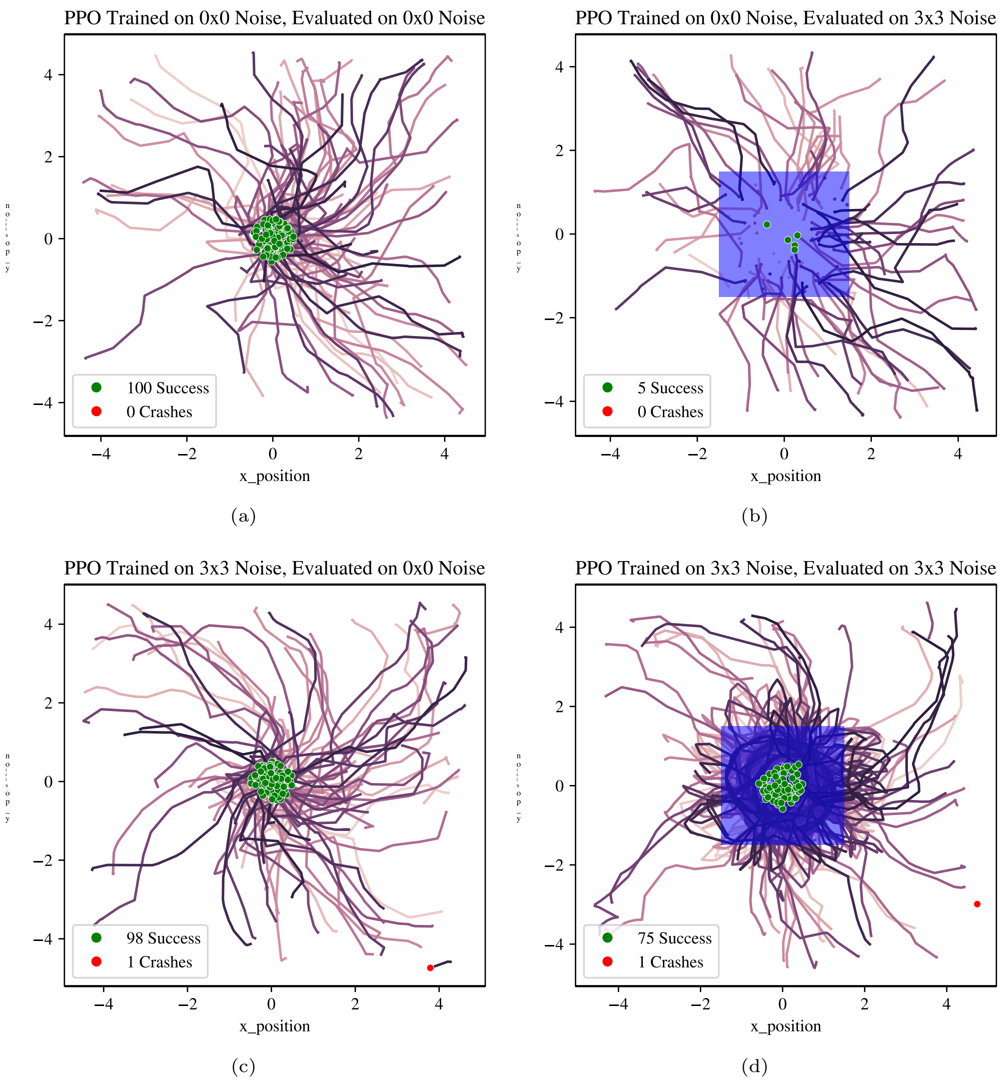


Fig. 19 在 0x0 地图 **(a)** 上评估的普通策略 (PPO 0x0) 评估期间生成的 100 个路径图，在 3x3 传感器拒绝地图 **(b)** 上评估的普通策略 (PPO 0x0)，在 0x0 传感器拒绝地图 **(c)** 上评估的对抗训练 (PPO 3x3)，以及在 3x3 传感器拒绝地图 **(d)** 上评估的对抗训练 (PPO 3x3)。红点表示碰撞，绿点表示成功导航至目标。蓝色区域显示传感器拒绝区域。

传感器禁区、静态目标位于中心且不存在任何障碍物）。图 19(a) 显示了在 0x0 图上评估的普通 PPO。在图 19(b) 中球门周围具有静态 3x3 噪声区域的地图上评估相同的普通策略。该策略无法实现目标，并且表现显着下降（从 100 次成功减少到 5 次成功）。当机器人进入噪声区域时，摄像头失效，

and the robot stops moving. This can be seen as many paths end inside the blue noise zone and do not continue.

In contrast, when adversarially trained with 3x3 random areas present in the environment and evaluated on the 3x3 map as shown in Fig. 19(d), the number of successful episodes increases to 75. When inside the blue sensor denied area, the agent simply moves in circles, as can be seen by path traces in the blue zone. This is not a reliable way of navigating with obstacles present in the environment but leads to success when no obstacles are present in the zone of sensor denial. The policy exploits the reward structure of the scenario, opting for a high-risk strategy of moving through the sensor denied areas. Hence, like the Lidar examples shown in Fig. 11, the policies have learned a high-risk strategy of blindly navigating when sensor readings are uncertain.

Since we are evaluating on the simplified map, no obstacles are present. Clearly, once in a sensor-denied area, the policies have no way of avoiding obstacles and this is deliberately not reflected in these results, in order to understand the policy behaviour in sensor-denied zones. Still, because the models were trained on the default map, they have learned to take the risky actions when in sensor-denied zones with obstacles present. This likely means that due to the reward shaping of the environment, it was more rewarding for the policy to take the high risk actions. However this only holds up with small amounts of sensor denied zones in the environment, as shown in Fig. 18, as larger amounts of noise result in failure to learn basic navigation behaviour.

3.7 Discussion

The overall best performer for the camera navigation was DreamerV3. DreamerV3 consistently outperformed other models for the camera observations for both 0x0 and 3x3 noise area evaluations, when trained on both 0x0 and 3x3 environments (Fig. 14 and 15). This might result from the DreamerV3 having a different structure to the other models and building a world model due to the autoencoder structure. We hypothesized that it might be able to overcome the sensor denied areas due to the autoencoder/recurrent structure. While it did outperform other models when trained and evaluated on the 3x3 environment (it reached a mean of 80.6 % in Fig. 14) that still does not match the baseline performance of 90.8% when trained and evaluated on the no noise environments. DreamerV3 was also the only model to have learned to navigate to a random goal using the camera-only observation space within 500,000 steps, as shown in section 3.4. Other models either required the Lidar policy (which contained information about distance and angle to target), or required a static goal to be able to converge to a solution using the camera observation space. DreamerV3 was also convenient to use as it did not require any hyperparameter tuning. The best performer (between TD3 and PPO) for the Lidar modality was TD3. The 0x0 trained TD3 model generally outperformed most variations on PPO across different evaluations.

In sections 3.2 and 3.6 we have learned that training policies with some noise present in the environment present an interesting result: these policies are more successful at navigating to the goal, but also take more high-risk actions when sensor readings are uncertain or completely denied, which may lead to colliding with obstacles more often - a terminating event in our scenario. This is a limitation of training

然后机器人停止移动。这可以看作是许多路径在蓝色噪声区域内结束并且不再继续。

相比之下，当使用环境中存在的 3x3 随机区域进行对抗性训练并在 3x3 地图上进行评估（如图 19（d）所示）时，成功的次数增加到 75。当在蓝色传感器拒绝区域内时，智能体只是简单地绕圈移动，如蓝色区域中的路径轨迹所示。这不是一种在环境中存在障碍物时可靠的导航方式，但当传感器拒绝区域中不存在障碍物时，这种方式会成功。该策略利用场景的奖励结构，选择穿过传感器拒绝区域的高风险策略。因此，就像图 11 中所示的激光雷达示例一样，这些政策已经学会了在传感器读数不确定时盲目导航的高风险策略。

由于我们正在简化地图上进行评估，因此不存在任何障碍。显然，一旦进入传感器拒绝区域，政策就无法避免障碍，并且故意不在这些结果中反映出来，以便了解传感器拒绝区域中的政策行为。尽管如此，由于模型是在默认地图上进行训练的，因此它们已经学会了在存在障碍物的传感器拒绝区域时采取危险的行动。这可能意味着，由于环境的奖励塑造，政策采取高风险行动会获得更大的奖励。然而，这只适用于环境中少量的传感器拒绝区域，如图 18 所示，因为大量的噪声会导致无法学习基本的导航行为。

3.7 Discussion

相机导航方面总体表现最佳的是 DreamerV3。当在 0x0 和 3x3 环境上进行训练时，DreamerV3 在 0x0 和 3x3 噪声区域评估的相机观察方面始终优于其他模型（图 14 和 15）。这可能是由于 DreamerV3 具有与其他模型不同的结构，并且由于自动编码器结构而构建了世界模型。我们假设它可能能够克服由于自动编码器/循环结构而导致的传感器拒绝区域。虽然它在 3x3 环境中训练和评估时确实优于其他模型（图 14 中的平均值达到 80.6%），但在无噪声环境中训练和评估时仍然不符合 90.8% 的基线性能。 DreamerV3 也是唯一一个学会使用仅相机观察空间在 500,000 步内导航到随机目标的模型，如第 3.4 节所示。其他模型要么需要激光雷达策略（其中包含有关目标距离和角度的信息），要么需要静态目标才能使用相机观察空间收敛到解决方案。 DreamerV3 使用起来也很方便，因为它不需要任何超参数调整。激光雷达模式的最佳性能（介于 TD3 和 PPO 之间）是 TD3。在不同的评估中，0x0 训练的 TD3 模型通常优于 PPO 的大多数变体。

在第 3.2 节和第 3.6 节中，我们了解到，环境中存在一些噪声的训练策略会带来一个有趣的结果：这些策略在实现目标方面更成功，但当传感器读数不确定或完全否认时，也会采取更多高风险的行动，这可能会导致更频繁地与障碍物相撞——在我们的场景中这是一个终止事件。这是训练的局限性

RL policies for safety-critical scenarios. Training them to maximize the reward and reach the goal in uncertain scenarios may lead to the policies learning to select actions that are deemed rewarding and may lead to achieving a reward, even if these actions may lead to a catastrophic scenario.

Korkmaz [31] reports that adversarially trained policies are more vulnerable to shifts in e.g. brightness or blurring than vanilla trained ones. While we did not consider adversarial learning algorithms (Korkmaz used SA-DDQN which is based on the SA-MDP and RADIAL algorithms, while we only trained our models on environments with adversarial perturbations), this could be mapped to our problem in that our noise-trained policies (e.g. 3x3 PPO) result in a larger amount of crashes and are less robust on the 0x0 evaluation, compared with vanilla PPO (0x0).

Mirowski et al. [6] perform a similar study which includes static mazes and large random mazes. These are equivalent to static goal (that we performed using camera), and random goal (that we performed using Lidar). In their paper, algorithms that used recurrence outperformed non-recurrent algorithms. We did not see such a trend in our results, and PPO-LSTM generally performed poorly (ignoring DreamerV3 which is also recurrent). This is surprising given that navigation is a partially observable Markov decision process (MDP), and hence, policies should benefit from memory.

4 Conclusion and Further Work

This study has shown quantitative results of how common reinforcement learning algorithms perform with different modalities - camera and Lidar - on the *DRL-Robot-Navigation* environment, and the different effects of training and evaluating on environments containing noisy and sensor denied areas. We observed that training DRL models on noisy environments creates policies that, generally perform better on noisy environments, compared with models trained on vanilla, no noise environments. However, we found that these policies opted for riskier actions when presented with a noisy or faulty observation: ones that might result in successful navigation to the goal, but also in collision with another object. While these models result in better performance by quantitative metrics, this behaviour is unacceptable from the point of view of safety-critical scenarios and shows a need for more robust RL methods that are resilient to noise or sensor outages in the scenarios. Hence, we open-source the environments that we built on top of the *DRL-Robot-Navigation* to enable other researchers to evaluate their model performance on these scenarios. The results presented here should serve as a benchmark for any future studies on strategies of how to deal with faulty sensors. While we present evaluation for Gaussian noise to Lidar and camera outages, different perturbations also should be integrated (e.g. [32] investigates transformations such as compression artefacts, brightness and contrast, and blur for camera observations). We consider that work on autoencoders could lead to identifying sensor failures, and in potentially serving as a backup that is able to fill out the gaps in case of sensor outages. Sensor fusion strategies should also be investigated. Hierarchical reinforcement learning could lead to better performance (as proposed by Havens et al. [33]) as it allows for the top hierarchy to detect adversaries. This could lead to

针对安全关键场景的强化学习策略。训练它们在不确定的情况下最大化奖励并实现目标可能会导致策略学习选择被视为奖励的行动并可能导致实现奖励，即使这些行动可能会导致灾难性的情况。

Korkmaz [31] 报告称，经过对抗性训练的政策更容易受到诸如以下方面的变化的影响：比普通训练过的亮度或模糊度更高。虽然我们没有考虑对抗性学习算法（Korkmaz 使用基于 SA-MDP 和 RADIAL 算法的 SA-DDQN，而我们只在具有对抗性扰动的环境中训练我们的模型），但这可以映射到我们的问题，因为与普通 PPO (0x0) 相比，我们的噪声训练策略（例如 3x3 PPO）会导致大量崩溃，并且在 0x0 评估上的稳健性较差。

米罗斯基等人。 [6]进行了一项类似的研究，其中包括静态迷宫和大型随机迷宫。这些相当于静态目标（我们使用相机执行）和随机目标（我们使用激光雷达执行）。在他们的论文中，使用递归的算法优于非循环算法。我们在结果中没有看到这样的趋势，并且 PPO-LSTM 通常表现不佳（忽略同样经常出现的 DreamerV3）。这是令人惊讶的，因为导航是部分可观察的马尔可夫决策过程（MDP），因此，策略应该受益于记忆。

4 Conclusion and Further Work

这项研究展示了常见强化学习算法如何在 *DRL- Robot-Navigation* 环境中以不同方式（相机和激光雷达）执行的定量结果，以及训练和评估对包含噪声和传感器遮挡区域的环境的不同影响。我们观察到，与在普通无噪声环境中训练的模型相比，在噪声环境中训练 DRL 模型创建的策略通常在噪声环境中表现更好。然而，我们发现，当出现嘈杂或错误的观察时，这些策略会选择风险更大的行动：这些行动可能会成功导航到目标，但也会与另一个物体发生碰撞。虽然这些模型通过定量指标获得了更好的性能，但从安全关键场景的角度来看，这种行为是不可接受的，并且表明需要更强大的强化学习方法，以适应场景中的噪声或传感器中断。因此，我们开源了在 *DRL-Robot-Navigation* 之上构建的环境，以使其他研究人员能够评估他们在这些场景下的模型性能。这里给出的结果应该作为未来有关如何处理故障传感器策略的任何研究的基准。虽然我们对激光雷达的高斯噪声和相机中断进行了评估，但也应该集成不同的扰动（例如[32]研究诸如压缩伪影、亮度和对比度以及相机观察的模糊等变换）。我们认为，自动编码器的工作可能会导致识别传感器故障，并有可能作为备份，能够在传感器中断的情况下填补空白。还应该研究传感器融合策略。分层强化学习可以带来更好的性能（正如 Havens 等人提出的[33]），因为它允许顶层检测对手。这可能会导致

better performance in tasks with noise, as the local goals might change through the course of the episode - from navigating to a target to getting out of noise.

Finally, we suggest further developments to the environment, to include environmental interactions, along with more elaborate models of sensor attacks.

Appendix A Learning Rate Sensitivity

As the algorithms (apart from DreamerV3) are sensitive to learning rate, we initially perform a learning rate study - we then continue using this learning rate throughout.

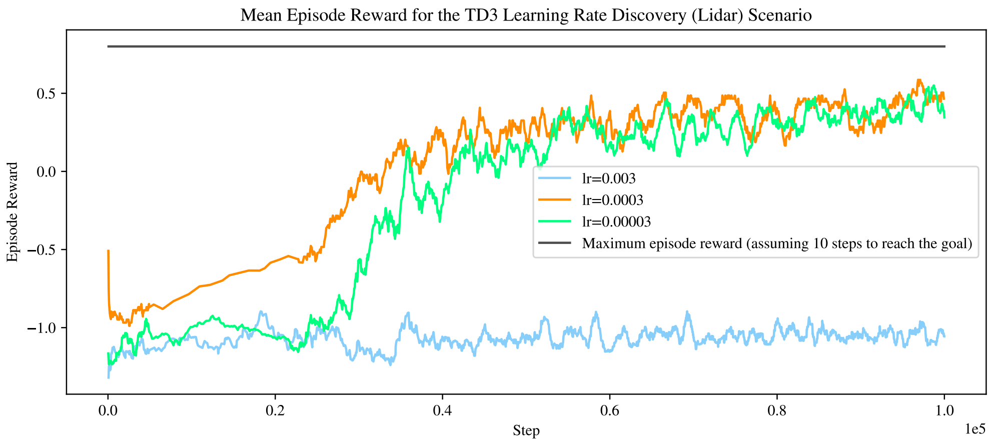


Fig. A1 TD3 Learning Rate Discovery (Lidar).

Fig. A1 shows the learning rate discovery for the TD3 algorithm using the Lidar observation space. Both, 0.0003 and 0.00003 values converge to approximately a max of 0.5 episode reward, but 0.0003 appears to converge more quickly.

Fig. A2 shows the learning rate discovery for the PPO algorithm using the Lidar observation space. A value of 0.003 appears to be the optimal value converging quicker than 0.0003.

Fig. A3 shows the learning rate discovery for the PPO LSTM algorithm using the Lidar observation space. No clear convergence was achieved - this is surprising because we expected recurrence networks to work on this problem. PPO without recurrence performs much better with the Lidar observation space as shown in Fig. A2.

Fig. A4 shows the learning rate discovery for the PPO LSTM algorithm using the camera observation space. A value of 0.0003 is the only one to converge to a solution. We will use this value of 0.0003 for both PPO and PPO LSTM camera runs.

Supplementary information. The modified *DRL-Robot-Navigation* environment is available https://github.com/mazqtpopx/cranfield-navigation-gym.

在有噪音的任务中表现更好，因为局部目标可能会在情节的过程中发生变化——从导航到目标到摆脱噪音。

最后，我们建议对环境进行进一步的开发，包括环境交互以及更复杂的传感器攻击模型。

Appendix A　Learning Rate Sensitivity

由于算法（DreamerV3 除外）对学习率敏感，因此我们最初执行学习率研究 - 然后我们继续使用该学习率。

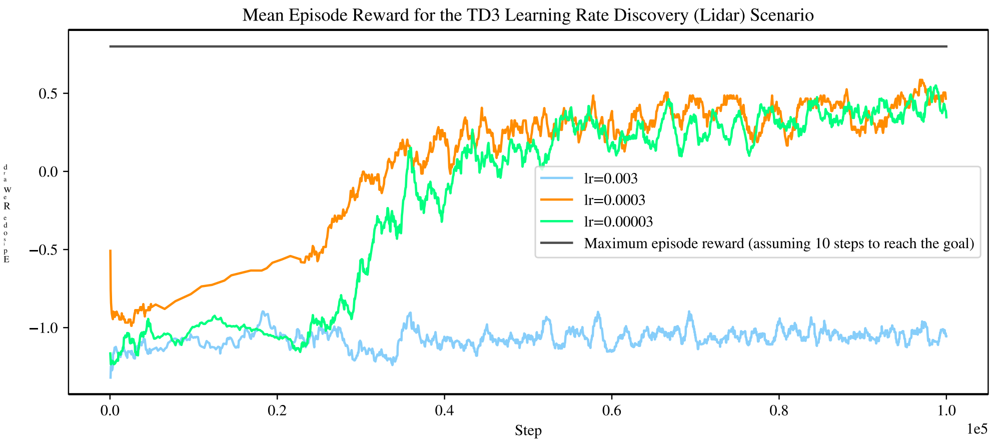


Fig. A1　TD3 学习率发现（激光雷达）。

图 A1 显示了使用激光雷达观测空间的 TD3 算法的学习率发现。 0.0003 和 0.00003 值都收敛到大约 0.5 集奖励的最大值，但 0.0003 似乎收敛得更快。

图 A2 显示了使用激光雷达观测空间的 PPO 算法的学习率发现。值 0.003 似乎是比 0.0003 收敛得更快的最佳值。

图 A3 显示了使用激光雷达观测空间的 PPO LSTM 算法的学习率发现。没有实现明显的收敛——这令人惊讶，因为我们期望递归网络能够解决这个问题。没有复发的 PPO 在激光雷达观测空间中表现得更好，如图 A2 所示。

图 A4 显示了使用相机观察空间的 PPO LSTM 算法的学习率发现。值 0.0003 是唯一收敛到解的值。我们将在 PPO 和 PPO LSTM 相机运行中使用这个值 0.0003。

Supplementary information.　修改后的 *DRL-Robot-Navigation* 环境可在 https://github.com/mazqtpopx/cranfield-navigation-gym 获取。

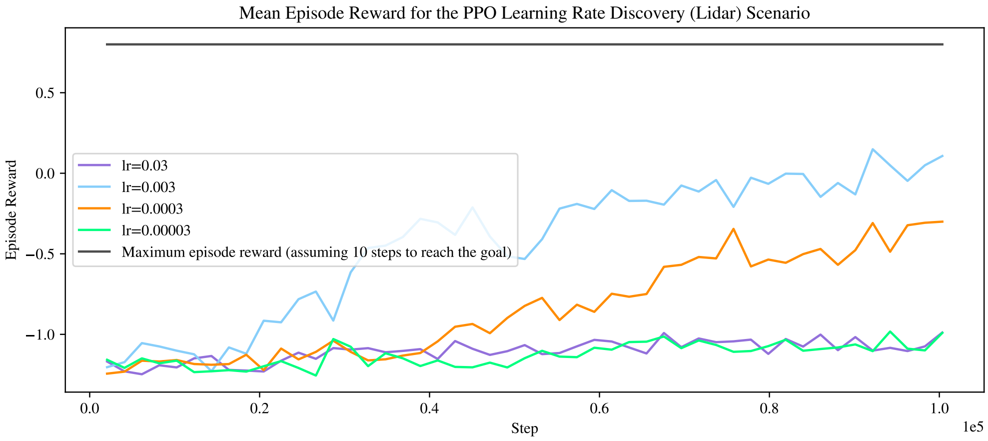


Fig. A2 PPO Learning Rate Discovery (Lidar).

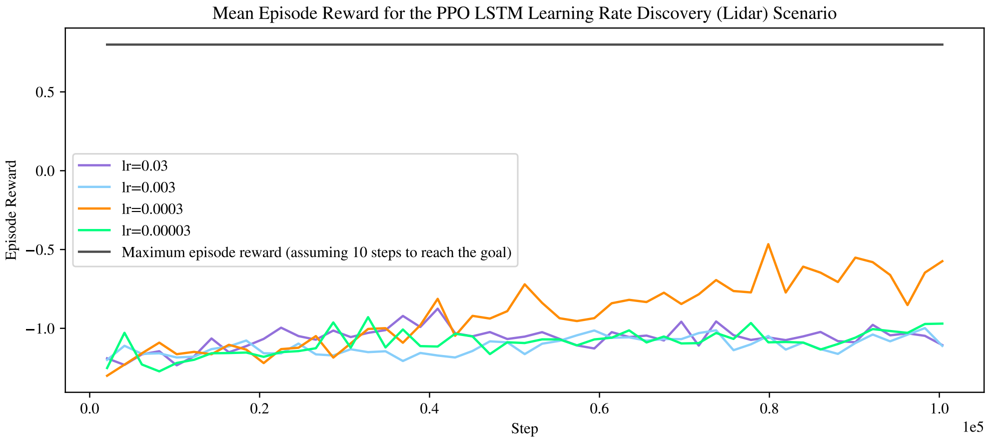


Fig. A3 PPO LSTM Learning Rate Discovery (Lidar).

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Declarations

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- Conflict of interest/Competing interests N/A
- Ethics approval and consent to participate: N/A

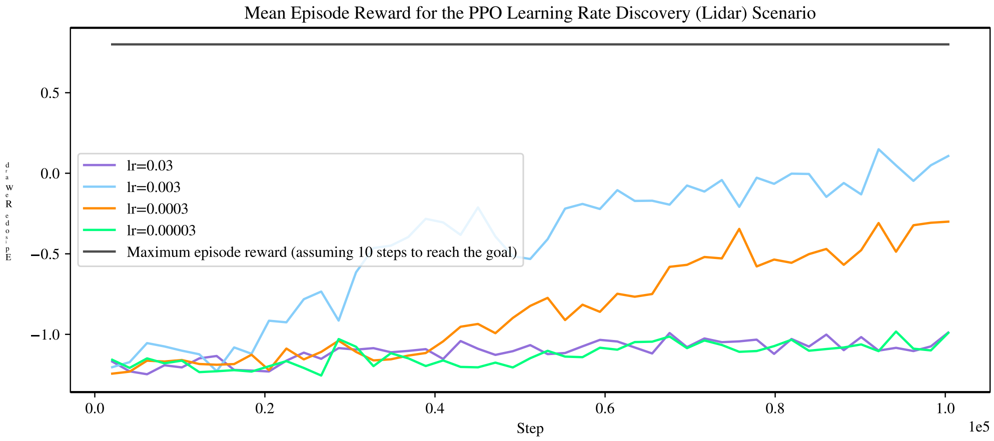

Fig. A2 PPO 学习率发现（激光雷达）。

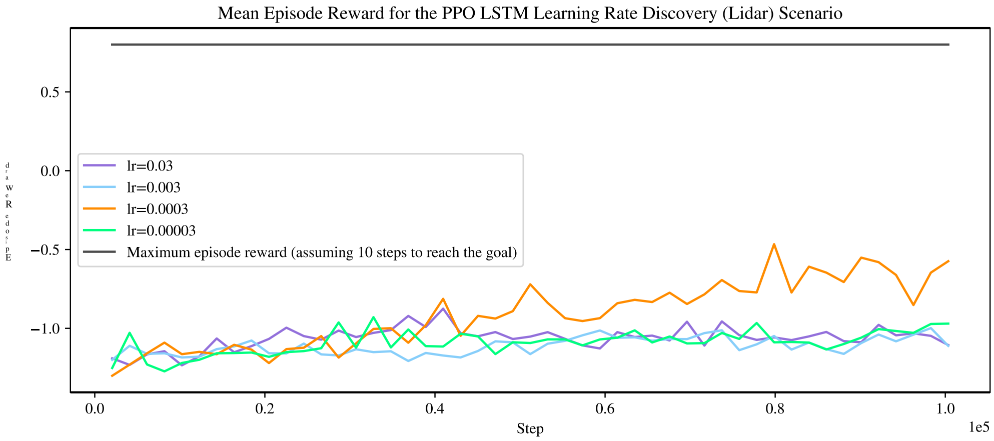

Fig. A3 PPO LSTM 学习率发现（激光雷达）。

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Declarations

- 资助：这项与 MW 和 PC 相关的工作得到了 Leonardo UK 和克兰菲尔德大学的支持，WG 和 AT 得到了 EPSRC TAS-S：值得信赖的自治系统：安全性 (EP/V026763/1) 的支持。
- 利益冲突/利益竞争 不适用
- 道德批准和同意参与：不适用

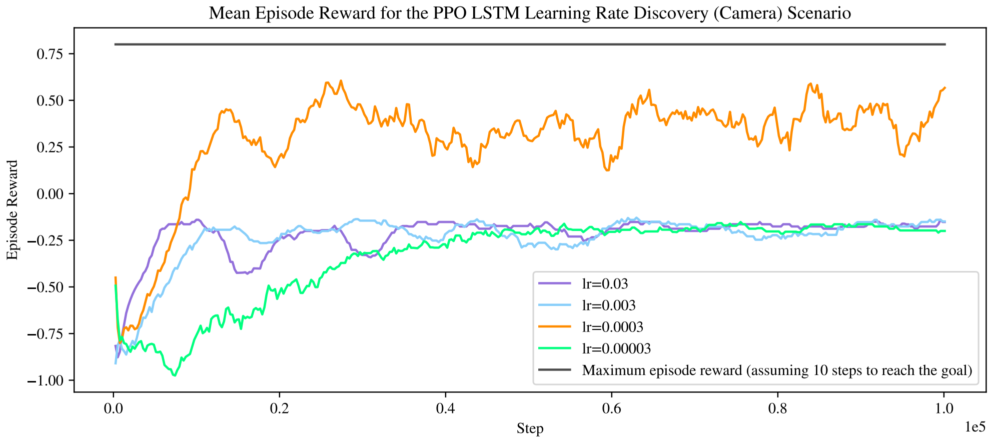

Fig. A4 PPO LSTM Learning Rate Discovery (Camera).

- Consent for publication: N/A
- Data availability: N/A
- Materials availability: N/A
- Code availability: code is available https://github.com/mazqtpopx/cranfield-navigation-gym.
- Author contribution: All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by MW, and PC. The first draft of the manuscript was written by MW and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

References

[1] Krizhevsky, A., Sutskever, I., Hinton, G.E.: ImageNet Classification with Deep Convolutional Neural Networks. In: Advances in Neural Information Processing Systems, vol. 25. Curran Associates, Inc. (2012). https://papers.nips.cc/paper_files/paper/2012/hash/c399862d3b9d6b76c8436e924a68c45b-Abstract.html Accessed 2023-08-01

[2] Deng, J., Dong, W., Socher, R., Li, L.-J., Kai Li, Li Fei-Fei: ImageNet: A large-scale hierarchical image database. In: 2009 IEEE Conference on Computer Vision and Pattern Recognition, pp. 248–255. IEEE, Miami, FL (2009). https://doi.org/10.1109/CVPR.2009.5206848 . https://ieeexplore.ieee.org/document/5206848/ Accessed 2021-07-21

[3] He, K., Zhang, X., Ren, S., Sun, J.: Deep Residual Learning for Image Recognition. arXiv:1512.03385 [cs] (2015). arXiv: 1512.03385. Accessed 2021-07-10

[4] Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A.A., Veness, J., Bellemare, M.G.,

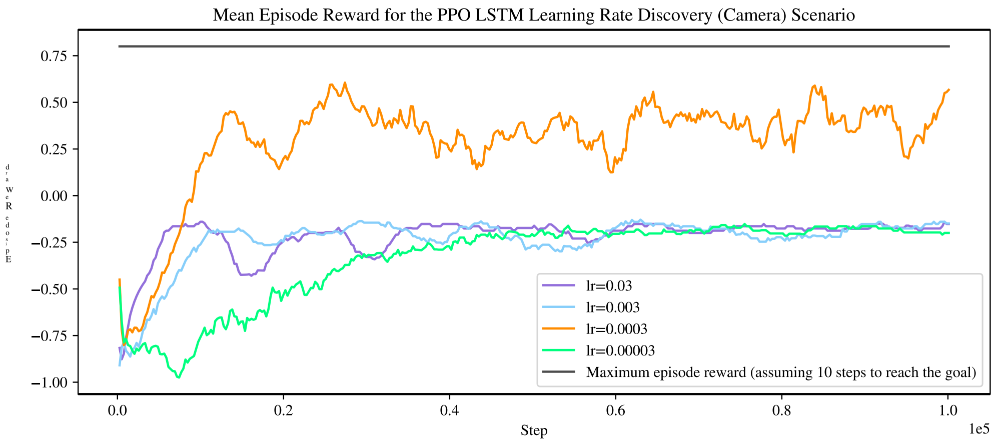

Fig. A4 PPO LSTM 学习率发现（相机）。

- 同意发表：不适用
- 数据可用性：不适用
- 材料可用性：不适用
- 代码可用性：代码可用 https://github.com/mazqtpopx/cranfield-navigation-gym。

- 作者贡献：所有作者都对研究构思和设计做出了贡献。通过MW和PC进行材料准备、数据收集和分析。该手稿的初稿由 MW 撰写，所有作者都对该手稿的先前版本发表了评论。所有作者阅读并批准了最终手稿。

References

[1] Krizhevsky, A.、Sutskever, I.、Hinton, G.E.：使用深度卷积神经网络进行 ImageNet 分类。见：神经信息处理系统的进展，卷。25. Curran Associates, Inc. (2012)。https://papers.nips.cc/paper_files/paper/2012/hash/c399862d3b9d6b76c8436e924a68c45b-Abstract.html 访问时间：2023-08-01 [2] Deng, J., Dong, W., Socher, R., Li, L.-J., Kai Li, Li Fei-Fei：ImageNet：大规模分层图像数据库。见：2009 年 IEEE 计算机视觉和模式识别会议，第 248-255 页。IEEE，佛罗里达州迈阿密 (2009)。https://doi.org/10.1109/CVPR.2009.5206848。https://ieeexplore.ieee.org/document/5206848/ 访问时间：2021-07-21 [3] He, K.、Zhang, X.、Ren, S.、Sun, J.：图像识别的深度残差学习。arXiv：1512.03385 [cs]（2015）。arXiv：1512.03385。访问时间：2021 年 7 月 10 日

[4] 姆尼赫，.V Kavukcuoglu, K.、Silver, D.、Rusu, A.A.、Veness, J.、B埃勒马尔，M.G.,

Graves, A., Riedmiller, M., Fidjeland, A.K., Ostrovski, G., Petersen, S., Beattie, C., Sadik, A., Antonoglou, I., King, H., Kumaran, D., Wierstra, D., Legg, S., Hassabis, D.: Human-level control through deep reinforcement learning. Nature **518**(7540), 529–533 (2015) https://doi.org/10.1038/nature14236 . Accessed 2022-12-10

[5] Walker, O., Vanegas, F., Gonzalez, F., Koenig, S.: A deep reinforcement learning framework for uav navigation in indoor environments. In: IEEE Aerospace Conference, pp. 1–14 (2019). https://doi.org/10.1109/AERO.2019.8742226

[6] Mirowski, P., Pascanu, R., Viola, F., Soyer, H., Ballard, A.J., Banino, A., Denil, M., Goroshin, R., Sifre, L., Kavukcuoglu, K., Kumaran, D., Hadsell, R.: Learning to Navigate in Complex Environments. arXiv. arXiv:1611.03673 [cs] (2017). http://arxiv.org/abs/1611.03673 Accessed 2022-10-17

[7] Cimurs, R., Suh, I.H., Lee, J.H.: Goal-Driven Autonomous Exploration Through Deep Reinforcement Learning. arXiv. arXiv:2103.07119 [cs] (2021). http://arxiv.org/abs/2103.07119 Accessed 2023-10-19

[8] Wisniewski, M., Ona, I.G., Chatzithanos, P., Guo, W., Tsourdos, A.: Autonomous Navigation in Dynamic Maze Environments Under Adversarial Sensor Attack. In: 2024 International Conference on Unmanned Aircraft Systems (ICUAS), pp. 881–886 (2024). https://doi.org/10.1109/ICUAS60882.2024.10557088 . ISSN: 2575-7296. https://ieeexplore.ieee.org/document/10557088/?arnumber=10557088 Accessed 2024-08-23

[9] Zhu, K., Zhang, T.: Deep reinforcement learning based mobile robot navigation: A review. Tsinghua Science and Technology **26**(5), 674–691 (2021) https://doi.org/10.26599/TST.2021.9010012 . Conference Name: Tsinghua Science and Technology. Accessed 2024-08-27

[10] Beattie, C., Leibo, J.Z., Teplyashin, D., Ward, T., Wainwright, M., Küttler, H., Lefrancq, A., Green, S., Valdés, V., Sadik, A., Schrittwieser, J., Anderson, K., York, S., Cant, M., Cain, A., Bolton, A., Gaffney, S., King, H., Hassabis, D., Legg, S., Petersen, S.: DeepMind Lab. arXiv. arXiv:1612.03801 [cs] (2016). http://arxiv.org/abs/1612.03801 Accessed 2023-05-19

[11] Kempka, M., Wydmuch, M., Runc, G., Toczek, J., Jaśkowski, W.: ViZDoom: A Doom-based AI Research Platform for Visual Reinforcement Learning. arXiv. arXiv:1605.02097 [cs] (2016). http://arxiv.org/abs/1605.02097 Accessed 2022-12-10

[12] Mnih, V., Badia, A.P., Mirza, M., Graves, A., Lillicrap, T.P., Harley, T., Silver, D., Kavukcuoglu, K.: Asynchronous Methods for Deep Reinforcement Learning. arXiv. arXiv:1602.01783 [cs] (2016). http://arxiv.org/abs/1602.01783 Accessed 2022-10-24

Graves, A.、Riedmiller, M.、Fidjeland, A.K.、Ostrovski, G.、Petersen, S.、Beattie, C.、Sadik, A.、Antonoglou, I.、King, H.、Kumaran, D.、Wierstra, D.、Legg, S.、Hassabis, D.：通过深度强化学习实现人类水平的控制。自然 **518**(7540)，529–533 (2015) https://doi.org/10.1038/nature14236 。访问时间：2022 年 12 月 10 日

[5] Walker, O., Vanegas, F., Gonzalez, F., Koenig, S.：室内环境中无人机导航的深度强化学习框架。见：IEEE 航空航天会议，第 1-14 页（2019 年）。 https://doi.org/10.1109/AERO.2019.8742226

[6] Mirowski, P.、Pascanu, R.、Viola, F.、Soyer, H.、Ballard, A.J.、Banino, A.、Denil, M.、Goroshin, R.、Sifre, L.、Kavukcuoglu, K.、Kumaran, D.、Hadsell, R.：学习在复杂环境中导航。 arXiv. arXiv：1611.03673 [cs]（2017）。 http://arxiv.org/abs/1611.03673 访问时间：2022 年 10 月 17 日

[7] Cimurs, R.、Suh, I.H.、Lee, J.H.：通过深度强化学习进行目标驱动的自主探索。 arXiv. arXiv：2103.07119 [cs]（2021）。 http://arxiv. org/abs/2103.07119 访问时间：2023-10-19

[8] Wisniewski, M.、Ona, I.G.、Chatzithanos, P.、Guo, W.、Tsourdos, A.：对抗性传感器攻击下动态迷宫环境中的自主导航。见：2024 年无人机系统国际会议 (ICUAS), 第 881–886 页 (2024)。 https://doi.org/10.1109/ICUAS60882.2024.10557088。国际标准期刊号：2575-7296。
https://ieeexplore.ieee.org/document/10557088/?arnumber=10557088 访问时间：2024 年 8 月 23 日

[9] 朱凯，张涛：基于深度强化学习的移动机器人导航：综述。清华科技 **26**(5), 674–691 (2021) https://do i.org/10.26599/TST.2021.9010012 。会议名称：清华科技。访问时间：2024年8月27日

[10] Beattie, C.、Leibo, J.Z.、Teplyashin, D.、Ward, T.、Wainwright, M.、Küttler, H.、Lefrancq, A.、Green, S.、Valdés, V.、Sadik, A.、Schrittwieser, J.、Anderson, K.、York, S.、Cant, M.、Cain, A.， Bolton, A.、Gaffney, S.、King, H.、Hassabis, D.、Legg, S.、Petersen, S.：DeepMind 实验室。 arXiv. arXiv：1612.03801 [cs]（2016）。 http://arxiv.org/abs/1612.03801 访问时间：2023-05-19

[11] Kempka, M.、Wydmuch, M.、Runc, G.、Toczek, J.、Jaśkowski, W.：ViZDoom：基于 Doom 的视觉强化学习人工智能研究平台。 arXiv. arXiv：1605.02097 [cs]（2016）。 http://arxiv.org/abs/1605.02097 访问时间：2022 年 12 月 10 日

[12] Mnih, V.、Badia, A.P.、Mirza, M.、Graves, A.、Lillicrap, T.P.、Harley, T.、Silver, D.、Kavukcuoglu, K.：深度强化学习的异步方法。 arXiv. arXiv：1602.01783 [cs]（2016）。 http://arxiv.org/abs/1602.01783 访问时间：2022 年 10 月 24 日

[13] Kaufmann, E., Bauersfeld, L., Loquercio, A., Müller, M., Koltun, V., Scaramuzza, D.: Champion-level drone racing using deep reinforcement learning. Nature **620**(7976), 982–987 (2023) https://doi.org/10.1038/s41586-023-06419-4 . Number: 7976 Publisher: Nature Publishing Group. Accessed 2023-09-10

[14] Polvara, R., Patacchiola, M., Sharma, S., Wan, J., Manning, A., Sutton, R., Cangelosi, A.: Toward end-to-end control for uav autonomous landing via deep reinforcement learning. In: International Conference on Unmanned Aircraft Systems (ICUAS) (2018)

[15] Chen, J., Li, S.E., Tomizuka, M.: Interpretable end-to-end urban autonomous driving with latent deep reinforcement learning. IEEE Transactions on Intelligent Transportation Systems **23**(6) (2022)

[16] Pasukonis, J., Lillicrap, T., Hafner, D.: Evaluating Long-Term Memory in 3D Mazes. arXiv. arXiv:2210.13383 [cs] (2022). http://arxiv.org/abs/2210.13383 Accessed 2024-04-24

[17] Lample, G., Chaplot, D.S.: Playing FPS Games with Deep Reinforcement Learning. arXiv. arXiv:1609.05521 [cs] (2018). http://arxiv.org/abs/1609.05521 Accessed 2022-10-14

[18] Luo, C., Yang, S.X.: A bioinspired neural network for real-time concurrent map building and complete coverage robot navigation in unknown environments. IEEE Transactions on Neural Networks **19**(7), 1279–1298 (2008) https://doi.org/10.1109/TNN.2008.2000394

[19] al., M.C.: Robust flight navigation out of distribution with liquid neural networks. Science Robotics **8** (2023)

[20] Sajjadi, S., Bittick, J., Janabi-Sharifi, F., Mantegh, I.: A robust and adaptive sensor fusion approach for indoor uav localization. In: International Conference on Unmanned Aircraft Systems (ICUAS) (2023)

[21] Negru, S.A., Geragersian, P., Petrunin, I., Guo, W.: Resilient multi-sensor uav navigation with a hybrid federated fusion architecture. Sensors **24** (2023)

[22] Liu, C., Zhang, G., Guo, W., He, R.: Kalman prediction-based neighbor discovery and its effect on routing protocol in vehicular ad hoc networks. IEEE Transactions on Intelligent Transportation Systems **21**(1), 159–169 (2020) https://doi.org/10.1109/TITS.2018.2889923

[23] Bae, S., Han, D., Park, S.: A new approach to lidar and camera fusion for autonomous driving. In: IEEE International Conference on Artificial Intelligence in Information and Communication (2023)

[13] Kaufmann, E., Bauersfeld, L., Loquercio, A., Müller, M., Koltun, V., Scaramuzza, D.：使用深度强化学习的冠军级无人机竞赛。自然 **620**(7976)，982–987 (2023) https://doi.org/10.1038/s41586-023-06419-4 。编号：7976 出版商：自然出版集团。访问时间：2023年9月10日

[14] Polvara, R.、Patacchiola, M.、Sharma, S.、Wan, J.、Manning, A.、Sutton, R.、Cangelosi, A.：通过深度强化学习实现无人机自主着陆的端到端控制。见：国际无人机系统会议 (ICUAS) (2018)

[15] Chen, J., Li, S.E., Tomizuka, M.：具有潜在深度强化学习的可解释的端到端城市自动驾驶。 IEEE 智能交通系统汇刊 **23**(6)（2022）

[16] Pasukonis, J.、Lillicrap, T.、Hafner, D.：评估 3D 迷宫中的长期记忆。 arXiv. arXiv：2210.13383 [cs]（2022）。 http://arxiv.org/abs/2210.13383 访问时间：2024-04-24

[17] Lample, G., Chaplot, D.S.：用深度强化学习玩 FPS 游戏。 arXiv. arXiv：1609.05521 [cs]（2018）。 http://arxiv.org/abs/1609.05521 访问时间：2022 年 10 月 14 日

[18] 罗成，杨S.X.：一种用于未知环境下实时并发地图构建和全覆盖机器人导航的仿生神经网络。 IEEE 神经网络汇刊 **19**(7), 1279–1298 (2008) https://doi.org/10.1 109/TNN.2008.2000394

[19] al., M.C.：使用液体神经网络实现分布式鲁棒飞行导航。科学机器人 **8**（2023）

[20] Sajjadi, S.、Bittick, J.、Janabi-Sharifi, F.、Mantegh, I.：一种用于室内无人机定位的鲁棒自适应传感器融合方法。见：国际无人机系统会议 (ICUAS)（2023 年）

[21] Negru, S.A.、Geragersian, P.、Petrunin, I.、Guo, W.：采用混合联合融合架构的弹性多传感器无人机导航。传感器 **24**（2023）

[22] 刘成，张刚，郭伟，何瑞：基于卡尔曼预测的车载自组织网络邻居发现及其对路由协议的影响。 IEEE 智能交通系统汇刊 **21**(1), 159–169 (2020) https://doi.org/10 .1109/TITS.2018.2889923

[23] Bae, S.、Han, D.、Park, S.：自动驾驶激光雷达和摄像头融合的新方法。见：IEEE 国际信息与通信人工智能会议 (2023)

[24] Kacker, T., Perrusquia, A., Guo, W.: Multi-spectral fusion using generative adversarial networks for uav detection of wild fires. In: IEEE Int. Conference on AI in Information and Communication (2023)

[25] Spaan, M.T.J.: Partially Observable Markov Decision Processes. In: Wiering, M., Otterlo, M. (eds.) Reinforcement Learning: State-of-the-Art, pp. 387–414. Springer, Berlin, Heidelberg (2012). https://doi.org/10.1007/978-3-642-27645-3_12 . https://doi.org/10.1007/978-3-642-27645-3_12 Accessed 2024-10-12

[26] Osogami, T.: Robust partially observable Markov decision process. In: Proceedings of the 32nd International Conference on Machine Learning, pp. 106–115. PMLR (2015). ISSN: 1938-7228. https://proceedings.mlr.press/v37/osogami15.html Accessed 2024-10-12

[27] Tessler, C., Efroni, Y., Mannor, S.: Action Robust Reinforcement Learning and Applications in Continuous Control. In: Proceedings of the 36th International Conference on Machine Learning, pp. 6215–6224. PMLR, ??? (2019). ISSN: 2640-3498. https://proceedings.mlr.press/v97/tessler19a.html Accessed 2024-03-04

[28] Zhang, H., Chen, H., Xiao, C., Li, B., Liu, M., Boning, D., Hsieh, C.-J.: Robust Deep Reinforcement Learning against Adversarial Perturbations on State Observations. In: Advances in Neural Information Processing Systems, vol. 33, pp. 21024–21037. Curran Associates, Inc. (2020). https://proceedings.neurips.cc/paper_files/paper/2020/hash/f0eb6568ea114ba6e293f903c34d7488-Abstract.html Accessed 2024-10-09

[29] Pinto, L., Davidson, J., Sukthankar, R., Gupta, A.: Robust Adversarial Reinforcement Learning. arXiv. arXiv:1703.02702 [cs] (2017). http://arxiv.org/abs/1703.02702 Accessed 2024-09-30

[30] Korkmaz, E.: Adversarial Training Blocks Generalization in Neural Policies. International Conference on Learning Representation (ICLR) Robust and Reliable Machine Learning in the Real World Workshop, (2021)

[31] Korkmaz, E.: Adversarial Robust Deep Reinforcement Learning Requires Redefining Robustness. Proceedings of the AAAI Conference on Artificial Intelligence **37**(7), 8369–8377 (2023) https://doi.org/10.1609/aaai.v37i7.26009 . Accessed 2024-09-28

[32] Korkmaz, E.: Understanding and Diagnosing Deep Reinforcement Learning. arXiv. arXiv:2406.16979 [cs] (2024). http://arxiv.org/abs/2406.16979 Accessed 2024-09-28

[33] Havens, A.J., Jiang, Z., Sarkar, S.: Online Robust Policy Learning in the Presence of Unknown Adversaries. arXiv. arXiv:1807.06064 [cs, stat] (2018). http://arxiv.org/abs/1807.06064 Accessed 2024-10-09

[24]Kacker, T.、Perrusquia, A.、Guo, W.：使用生成对抗网络进行无人机野火检测的多光谱融合。见：IEEE 国际。信息与通信人工智能会议（2023）

[25] Spaan, M.T.J.：部分可观察的马尔可夫决策过程。见：Wiering, M.、Otterlo, M.（编辑）《强化学习：最先进的技术》，第 387-414 页。施普林格，柏林，海德堡（2012）。 https://doi.org/10.1007/978-3-642-27645-3 12 .
https://doi.org/10.1007/978-3-642-27645-3_12 访问时间：2024 年 10 月 12 日

[26] Osogami, T.：鲁棒部分可观察马尔可夫决策过程。见：第 32 届国际机器学习会议论文集，第 106-115 页。 PMLR（2015）。国际标准期刊号：1938-7228。
https://proceedings.mlr.press/v37/osogami15.html 访问时间：2024 年 10 月 12 日

[27] Tessler, C.、Efroni, Y.、Mannor, S.：动作鲁棒强化学习及其在连续控制中的应用。见：第 36 届国际机器学习会议论文集，第 6215-6224 页。 PMLR，？？？ （2019）。 ISSN：2640-3498。https://proceedings.mlr.press/v97/tessler19a.html 访问时间：2024 年 3 月 4 日

[28] 张 H.、陈 H.、肖 C.、李 B.、刘 M.、Boning, D.、谢 C.-J.：针对状态观测的对抗性扰动的鲁棒深度强化学习。见：神经信息处理系统的进展，卷。 33，第 21024–21037 页。 Curran Associates, Inc. (2020)。
https://proceedings.neurips.cc/paper_files/paper/2020/hash/f0eb6568ea114ba6e293f903c34d7488-Abstract.html 访问时间：2024 年 10 月 9 日

[29] Pinto, L.、Davidson, J.、Sukthankar, R.、Gupta, A.：鲁棒对抗性强化学习。 arXiv. arXiv：1703.02702 [cs]（2017）。 http://arxiv.org/abs/1703.02702 访问时间：2024-09-30

[30] Korkmaz, E.：神经策略中的对抗性训练块泛化。国际学习表征会议 (ICLR) 现实世界中鲁棒可靠的机器学习研讨会，（2021 年）

[31] Korkmaz, E.：对抗性鲁棒深度强化学习需要重新定义鲁棒性。 AAAI 人工智能会议论文集 **37**(7), 8369–8377 (2023) https://doi.org/10.1609/aaai.v37i7.26009 。访问时间：2024年9月28日

[32] Korkmaz, E.：理解和诊断深度强化学习。 arXiv. arXiv：2406.16979 [cs]（2024）。 http://arxiv.org/abs/2406.16979 访问时间：2024-09-28

[33] Havens, A.J.、Jiang, Z.、Sarkar, S.：未知对手存在下的在线稳健政策学习。 arXiv. arXiv：1807.06064 [cs，统计]（2018）。 http://arxiv. org/abs/1807.06064 访问时间 2024-10-09

[34] Panda, D.K., Guo, W.: Action Robust Reinforcement Learning for Air Mobility Deconfliction Against Conflict Induced Spoofing. IEEE Transactions on Intelligent Transportation Systems, 1–13 (2024) https://doi.org/10.1109/TITS.2024.3454354 . Conference Name: IEEE Transactions on Intelligent Transportation Systems. Accessed 2024-10-11

[35] Towers, M., Kwiatkowski, A., Terry, J., Balis, J.U., De Cola, G., Deleu, T., Goulão, M., Kallinteris, A., Krimmel, M., KG, A., Perez-Vicente, R., Pierré, A., Schulhoff, S., Tai, J.J., Tan, H., Younis, O.G.: Gymnasium: A Standard Interface for Reinforcement Learning Environments. arXiv. arXiv:2407.17032 [cs] (2024). http://arxiv.org/abs/2407.17032 Accessed 2024-08-27

[36] Kim, S.-G., Lee, E., Hong, I.-P., Yook, J.-G.: Review of Intentional Electromagnetic Interference on UAV Sensor Modules and Experimental Study. Sensors **22**(6), 2384 (2022) https://doi.org/10.3390/s22062384 . Number: 6 Publisher: Multidisciplinary Digital Publishing Institute. Accessed 2023-07-27

[37] Fujimoto, S., Hoof, H., Meger, D.: Addressing Function Approximation Error in Actor-Critic Methods. arXiv. arXiv:1802.09477 [cs, stat] (2018). http://arxiv.org/abs/1802.09477 Accessed 2024-02-18

[38] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., Klimov, O.: Proximal Policy Optimization Algorithms. arXiv. arXiv:1707.06347 [cs] (2017). http://arxiv.org/abs/1707.06347 Accessed 2022-11-03

[39] Hochreiter, S., Schmidhuber, J.: Long Short-Term Memory. Neural Computation **9**(8), 1735–1780 (1997) https://doi.org/10.1162/neco.1997.9.8.1735 . Accessed 2024-03-28

[40] Hafner, D., Pasukonis, J., Ba, J., Lillicrap, T.: Mastering Diverse Domains through World Models. arXiv. arXiv:2301.04104 [cs, stat] (2023). http://arxiv.org/abs/2301.04104 Accessed 2023-12-06

[41] Raffin, A., Hill, A., Gleave, A., Kanervisto, A., Ernestus, M., Dormann, N.: Stable-Baselines3: Reliable Reinforcement Learning Implementations. Journal of Machine Learning Research **22**(268), 1–8 (2021). Accessed 2024-10-09

[34] Panda，D.K.，Guo，W.：针对冲突引起的欺骗的空中机动去冲突的行动鲁棒强化学习。 IEEE 智能交通系统汇刊，1-13 (2024) https://doi.org/10.1109/TITS.202 4.3454354。会议名称：IEEE 智能交通系统汇刊。访问时间：2024年10月11日

[35] Towers, M.、 Kwiatkowski, A.、 Terry, J.、 Balis, J.U.、 De Cola, G.、 Deleu, T.、 Goulão, M.、 Kallinteris, A.、 Krimmel, M.、 KG, A.、 Perez-Vicente, R.、 Pierreé, A.、 Schulhoff, S.、 Tai, J.J.、 Tan， H., Younis, O.G.：Gymnasium：强化学习环境的标准界面。 arXiv. arXiv：2407.17032 [cs]（2024）。 http://arxiv.org/abs/2407.17032 访问时间：2024-08-27

[36] Kim, S.-G., Lee, E., Hong, I.-P., Yook, J.-G.：无人机传感器模块有意电磁干扰及实验研究综述。传感器 **22**(6)，2384 (2022) https://doi.org/10.3390/s22062384 。数量：6 出版商：多学科数字出版研究所。访问时间：2023年7月27日

[37] Fujimoto, S.、 Hoof, H.、 Meger, D.：解决 Actor-Critic 方法中的函数逼近错误。 arXiv. arXiv：1802.09477 [cs， 统计]（2018）。 http://arxiv.org/abs/1802.09477 访问时间：2024-02-18

[38] Schulman, J.、 Wolski, F.、 Dhariwal, P.、 Radford, A.、 Klimov, O.：近端策略优化算法。 arXiv. arXiv：1707.06347 [cs]（2017）。 http://arxiv.org/abs/1707.06347 访问时间：2022 年 11 月 3 日

[39] Hochreiter, S., Schmidhuber, J.：长短期记忆。神经计算 **9**(8)，1735–1780 (1997) https://doi.org/10.1162/neco.1997.9.8.1735 。访问时间：2024年3月28日

[40] Hafner, D.、 Pasukonis, J.、 Ba, J.、 Lillicrap, T.：通过世界模型掌握不同领域。 arXiv. arXiv:2301.04104 [cs, stat] (2023)。 http://arxiv.org/abs/2301.04104 访问时间：2023-12-06

[41] Raffin, A.、 Hill, A.、 Gleave, A.、 Kanervisto, A.、 Ernestus, M.、 Dormann, N.：稳定基线 3：可靠的强化学习实现。机器学习研究杂志 **22**(268)，1–8 (2021)。访问时间：2024年10月9日